

ABRA — Architecture Review

2026-09-09 — engineering review, systems audit and the pre-6.0.0 list; a dated record, not current state

Seven division agents, coordinated and verified by the session coordinator · 2026-09-09

Contents

1. ARCHITECTURE-REVIEW-2026-09-09

PART 1

ABRA — architecture review, 2026-09-09

Dated record, not current state. Every number in this document cites the command that produced it or the artifact field (`path:field`) it was read from. Do not take a number out of this file: `node engine/status.js` is the state, `node engine/open_work.js` is the open work.

- **Commit reviewed:** `d6789951` (2026-09-09 11:53 -0400, "Five failure shapes and six sources this session paid for, written where the next one already reads").
- **Produced by** seven read-only division agents (ENGINE, MEASURE, OPS, a code review, a test-mutation pass, a store-and-authority forensics pass, a causal-claims-and-boundary audit) plus one worktree mutation pass, coordinated and verified by the session coordinator. Where a claim below says "coordinator confirmed", the command and output are in the coordinator's verification log for this session; where it says "per report", the receipt is in the named report and was not independently re-run.
- **Serves three commands Will invoked today:** the CTO engineering review (verdict), the principal-architect systems audit (rules with originating failure and enforcing check), and the "everything to address before 6.o.o" list. It also carries the audit-report deliverable (findings, changes, corrected numbers, causal-claims audit, what could not be done).
- **Reports it draws on** (all dated 2026-09-09, all historical by construction):
 - `docs/_reports/2026-09-09-pre-600-engine-review.md`
 - `docs/_reports/2026-09-09-pre-600-measure-review.md`
 - `docs/_reports/2026-09-09-pre-600-ops-review.md`
 - `docs/_reports/2026-09-09-pre-600-code-review.md`
 - `docs/_reports/2026-09-09-pre-600-test-breaks.md`
 - `docs/_reports/2026-09-09-pre-600-store-and-authority.md`
 - `docs/_reports/2026-09-09-pre-600-causal-and-boundary.md`
 - `docs/_reports/2026-09-09-thesis-defence-notes.md` (companion to `docs/THESIS-DEFENCE-REVIEW-2026-09-09.md`, which this document links and does not duplicate)
- **Vocabulary, expanded once.** MEDICHAM is the in-house battle simulator (`engine/medicham2-browser.js`). "The authority" is the pinned checkout of Pokémon Showdown, the official simulator, at commit `20ad99ff` . "The gate" is the computed quarantine gate in `engine/status.js` / `engine/quarantine.js` that decides whether MEDICHAM is correct enough for anything downstream of it to be quoted. "The differential" is `engine/game_differential.js`, which plays the same games in both engines and compares them turn by turn; its artifact is `data/game-differential.json` . "The store" is the append-only corpus of real ladder games. "A release" is a frozen copy of the engine's source files cut by `engine/engine_release.js`, identified by a twelve-character digest. "The pinned pool" is `data/team-pool-frozen`, a frozen sample of real ladder teams. MAG, MILTANK, PORYGON2, DODUO and the R1–R4 series are models and measurements downstream of MEDICHAM; they are QUARANTINED and none of their figures is quoted here — the word "withheld" marks each place one would otherwise appear.

Executive summary

What was found. The headline claim 6.o.o would publish — "MEDICHAM agrees with Showdown on every board, 0 of 958 games" — is true of the 958 games it counts and is not yet a claim about the run. The run played 961 games. Three were removed by a rule that voids a game when the two engines' dice addresses stop agreeing, and every one of those three is a game whose boards parted (`data/game-differential.json:mid_void.void_game_tags[].board_parted_at_turn = 4, 6, 3`; coordinator confirmed). The counterexamples left the denominator before the ratio was taken. Separately, the corpus under every store-derived figure is missing 15,862 games: the 2026-09-06 sharding cutover cut shards from a local file two weeks behind the tracked one, and the notes row for that change reads "Supersedes. Nothing." — 11,110 ladder and 4,752 bo3 game ids are in the last committed monolith (`d2a418a5`) and in no tracked shard (coordinator confirmed by id-set difference). The release every 6.o.o artifact stamps, `b730e44f3314`, is not in the repository (0 tracked files under `data/releases/b730e44f3314`; 26 of 636 releases tracked; `.gitignore:174`). A scheduled GitHub job rewrites a frozen engine source on 2026-09-11 07:00 UTC. The documentation-currency gate cannot see a wrong headline number (a

mutation changing 27 of 961 to 41 of 961 in docs/MODELS.md passed byte-identically). The test suite is 150 passed / 27 failed / 0 skipped of 177 with no red waived by name, including a registered gate at exit 1. Five refuted causal claims stand in living documents, one in CLAUDE.md itself. The white paper states the opposite of three of its own artifacts. The thesis defence returned MAJOR REVISIONS with twelve required analyses.

What was fixed this session. Three of the findings above are closed, each with a receipt the coordinator re-ran. **The store** (finding 1): the 11,110 ladder and 4,752 bo3 games are back in five new write-once shards — tracked ids ladder 81,269 → 92,379, bo3 28,049 → 32,801, ots 4,167 unchanged; comm -23 of d2a418a5 's ids against all shards reads 0 for ladder and 0 for bo3 (coordinator confirmed, both stores); no existing shard changed. The check that would have caught it now exists: build/compress-stores.js --check asserts the shards carry every id git tracked at HEAD~1, shown red on a one-row break and red on a worktree replay of the cutover (LOST 11513 / LOST 5764). **The cron** (finding 5): smogon-stats.yml now writes data/smogon-priors.observed.json and restores the frozen source; workflow-only, so engine_release.js drift b730e44f3314 reads NO-DRIFT (coordinator confirmed). **The documents** (findings 7 and 8): five refuted causal claims rewritten across 8 locations, 22 figures withdrawn or replaced across 21 locations in 7 documents, and tests/test-docs-current.js went 32/1 → 33/0 (coordinator re-ran: 33 passed, 0 failed). The three white-paper sections now state what their artifacts say; the re-analyses the defence asked for (T1–T5) are still owed. This lands as MINOR 5.275.0 — 22 published figures moved under an unchanged basis. Every file, diff and test result is in §3; the corrected numbers are in §4.

What is still broken. Everything in §2 marked BLOCKS 6.0.0 that §3 does not close: the three void games need a per-game attribution (engine or instrument) before "0 of 958" can be printed as "correct"; the roster gate passes on 139 of 202 abilities compared; the narration clause is red at 12 of 961 across 13 causes; 27 reds; the docs gate's two blind spots; the release evidence chain.

What could not be verified. The stranded-release count across all 636 releases (the instrument dies at a 2–3 GB heap); whether engine/side_selection_census.js is red (code review says exit 1, coordinator observed exit 0); the cause of the residual trio; the share of living-document paragraphs the docs gate's QUALIFIED exemption removes; the thesis's twelve generators under the current quality filter (none re-run). Full list in §6.

VERDICT: FIX THEN SHIP.

This is not a rebuild candidate. The instruments that found every item above are this project's own — the void tags, the by-cause totals, the release stamps, the id sets in git history — and most of the fixes are hours. It is not shippable today because, if 6.0.0 ran as the tree stands and was wrong, the failure is not a crash. It is a published sentence, "MEDICHAM agrees with Showdown on every board", resting on a denominator that excluded its own three counterexamples, measured on a release nobody can re-open, over a corpus missing 15,862 games nobody knew were gone — a confidently wrong number wearing a receipt, which is the exact failure this project says it exists to prevent.

What the team must change to keep confidence (the pre-6.0.0 list in §8, condensed): restore the 15,862 games and add the id-superset check to the cutover; attribute the three void games and print "0 of 958 usable, 3 voided, N engine" in those words; track b730e44f3314 with git add -f; stop the smogon cron writing a frozen source before Thursday; make the docs gate compare the value at the claim; fix or waive by name every one of the 27 reds; strike the five refuted causal claims; rewrite the three white-paper sections from their artifacts' verdict fields; state the compared set (139 of 202) wherever the roster is cited; add the second publisher to the WHO MAY WRITE table.

Findings, ranked by blast radius

Ranking rule: what could put a wrong number in front of a reader first; untidiness last. Severity is one of BLOCKS 6.0.0 / SHOULD FIX BEFORE / DEFER WITH REASON. "Fixed" is stated only where a fix report or a coordinator command says so; otherwise "No".

1. The sharding cutover silently dropped 11,110 ladder and 4,752 bo3 games from the tracked store

- **What is wrong.** On 2026-09-06 the monolithic compressed stores were replaced by dated shards cut from the LOCAL plain file, which had diverged from origin's tracked copy for two weeks. The comparison that would have caught it — shards versus the tracked .gz they replaced — was never made; the cutover was verified against the live local file ("IDENTICAL 12 shard(s) - > 76833 rows", docs/_reports/2026-09-06-store-sharding-and-reclaim.md:41-56 per the forensics report) and the notes row (docs/RUNNING-NOTES.md:573 block) declares "**Supersedes.** Nothing. No published figure changed value."

- **How it was proved** (coordinator): `git show d2a418a5:data/games.ladder.jsonl.gz | gzip -dc | grep -o '"id": "[^"]*"' | sort -u | wc -l` → 88346; the same over `data/parsed/games.ladder/*.jsonl.gz` → 81269; `comm -23 mono.ids shard.ids | wc -l` → **11110** (in the monolith, in no shard); `comm -13` → 4033 (new since cutover); `bo3 comm -23` → **4752**. `git log -1 d2a418a5` → 2026-09-06 15:34:02 +0000 (an abra-bot ingest); `git log -1 18432bcb` → 2026-09-06 16:25:38 -0400 (the cutover, 51 minutes later).
- **What it could have put in front of a reader.** Every store-derived figure (usage, bring rates, `data/meta-usage.json`, the legality verdict, the quality funnel) computed on a corpus 15,862 games short, under a row that says nothing changed. The workflow's own shrink guard could not see it: its baseline is `wc -l` of the store restored from shards, so a shard set that starts short is the new floor (per the forensics report, `ingest.yml` "Refuse to proceed if a store shrank").
- **Recoverable.** History is permanent; `git show d2a418a5:...` holds every row. 3,510 of the 11,110 also have raw logs in tracked `data/raw/` shards; 7,600 exist only as parsed rows (per report).
- **Fixed? FIXED** (`docs/_reports/2026-09-09-fix-store-recovery.md`). Recovered from `d2a418a5` into five new shards, `data/parsed/games.ladder/20260909T1708-{00,01,02}.jsonl.gz` and `data/parsed/games.bo3/20260909T1708-{00,01}.jsonl.gz`; tracked ids ladder 81,269 → 92,379, bo3 28,049 → 32,801, ots 4,167 unchanged. Proof of closure: `comm -23` of `d2a418a5`'s ids against all shards → **0** ladder, **0** bo3 (coordinator re-ran both); `git diff --stat HEAD -- data/parsed` empty (no existing shard changed). The missing check landed too: `build/compress-stores.js --check` now asserts the shards $\supseteq$ the ids tracked at `HEAD~1` in either form — green on the tree in 8.7 s, red on a one-row sandbox break, red on a worktree replay of `18432bcb` (`LOST 11513 ladder / LOST 5764 bo3`). Still owed: wiring `--check` into `ingest.yml` (it runs from `tests/test-workflow-paths.js` only; the runner's shrink guard still measures against itself).
- **Severity: BLOCKS 6.o.o. Owner: OPS (recovery) and MEASURE (the missing superset check).**

2. The three games the void rule excluded are the three games whose boards parted

- **What is wrong.** `void_games_counted_against_the_engine: false` is a rule, not a measurement. A game whose boards part will have disagreeing dice addresses from that turn on, so the rule is structurally biased toward removing exactly the games the bar exists to count. "BOARD-MATERIAL o of 958" is the count after the counterexamples left the denominator.
- **How it was proved** (coordinator, `node -e` over `data/game-differential.json`): `state.games` 958, `state.games_board_never_diverged` 958, `state.games_void_excluded` 3; `mid_void.void_games` 3, `mid_void.usable_games` 958, `mid_void.diverged_among_usable` 13; `mid_void.void_game_tags` three entries, why "low-identity", `board_parted_at_turn` 4 / 6 / 3; `mid_void.by_reason` shared-addresses-agree 958 (diverged 13), low-identity 3 (diverged 3). Also `end_state[0].summary.by_cause_totals.games_board_material` = 3 (2 same turn, 1 parted later) and `verdicts` = SAME-END-STATE 958, DIFFERENT-END-STATE 2, THREW 1.
- **What the three are** (ENGINE report §1x, from `first_divergences`): (a) a charge move that announced one body and struck the other — turn 3, `omit-spread ...2658645239`; the authority KO'd the announced target, this engine wrote a resisted hit on its partner; Phantom Force, 1,701 pool slots, same release path as Solar Beam (15,867) and Electro Shot (9,600); (b) a Future Sight slot-condition end ordered against a Reflect side-condition end differently, board parted at turn 6 — `omit-spread ...2657358877`; (c) a post-switch Scorching Sands burn applied here and not there, turn 4 — `omit-protect ...2662758209`, the shape the session-close attributes to the INSTRUMENT (`freshBodies` dropping `_switchKey`, proven with a control on a different game). "Probably the instrument" is not an attribution.
- **What it could have put in front of a reader.** "o of 958" printed as "correct against Showdown" with two DIFFERENT-END-STATE games — one a KO versus a resisted hit, which is winner-shaped — sitting outside the count.
- **Fixed?** No. Owed: one probe per game and an attribution; then the sentence reads "o of 958 usable, 3 voided, of which N were the engine", with N derived.
- **Severity: BLOCKS 6.o.o. Owner: ENGINE (attribution), MEASURE (the rule's bias, stated in the artifact).**

3. The documentation-currency gate cannot see a wrong headline figure

- **What is wrong.** Two independent mechanisms in `engine/docs_scan.js`. (1) :626 `QUALIFIED` is a word list (`retract|withdraw|superseded|void|...|prior|former|previously|...`) and :807 skips the WHOLE paragraph if any word matches — `docs/MODELS.md:7` contains "superseded", so every figure in the ledger's headline paragraph is unchecked. (2) :496-499 `artifactHas` tests membership of the figure in the SET OF EVERY NUMBER in the cited artifact; `data/game-`

differential.json (471 KB) contains the token 41 on 5 lines and 27 on 7, so a wrong headline whose digits occur anywhere in the file passes.

- **How it was proved** (test-breaks report (h), in a worktree at d6789951): 27 of 961 → 41 of 961 in docs/MODELS.md:7 : exit 1, 7684 B / 105 lines, byte-identical to green, "no new entries (baseline 61, now 61)". 27 of 962 : same. A NEW paragraph with 777 of 961 : caught. The same new paragraph with 41 : not caught. The coordinator read the two source lines cited.
- **What it could have put in front of a reader.** Any wrong figure in any paragraph that mentions a retraction, and any wrong figure whose digits appear somewhere in a large artifact — with the gate green. MEASURE §9 adds a third, bounded gap: a figure whose only citations are data/*.js bundles is skipped entirely (:803 if (!sets.length) continue;); seven such lines in the living set.
- **Fixed?** No. The fix shape is to compare the value at the claim's named field, not membership in the artifact, and to exempt the SENTENCE that carries the qualifier, not the paragraph.
- **Severity: BLOCKS 6.o.o** (the gate that would catch every other document error in this list is blind to the class that matters). **Owner: MEASURE.**

4. The release every 6.0.0 artifact stamps is not in the repository

- **What is wrong.** data/game-differential.json, data/all-mechanics-fire.json and data/roster.{moves,items,abilities}.json all stamp engine_release: b730e44f3314 . That directory is untracked.
- **How it was proved** (coordinator): git ls-files data/releases | grep -c release.json → 26; ls data/releases | wc -l → 636; git ls-files data/releases/b730e44f3314 → nothing; .gitignore:174 data/releases/. The ignore block's own text (.gitignore:167-173) says to add a cited release deliberately with git add -f data/releases/<id> . It has not been.
- **What it could have put in front of a reader.** After a clone, or after this disk, the evidence chain for every 6.0.0 figure ends at a twelve-character id. CLAUDE.md calls a stamped release "the first thing in this repo that can be VERIFIED rather than assumed" — only if the bytes are in the repo. MEASURE §4 adds that the id is not reproducible on another machine either: nine of the 27 SOURCES are CRLF on disk and unpinned, so a fresh checkout digests differently (the read side is EOL-insensitive; the ID is not).
- **Fixed?** No. One command: git add -f data/releases/b730e44f3314 .
- **Severity: BLOCKS 6.o.o. Owner: MEASURE.**

5. A scheduled job rewrites a frozen engine source on 2026-09-11 07:00 UTC

- **What is wrong.** .github/workflows/smogon-stats.yml runs node engine/smogon_priors.js on a cron and commits data/smogon-priors.json . engine/smogon_priors.js:219 stamps generated: new Date().toISOString().slice(0,10) , so the bytes change on every run even with no new month. data/smogon-priors.json is a frozen SOURCE (engine/engine_release.js:150). Every release cut before the run drifts; every measurement on it becomes "engine moved since".
- **How it was proved** (coordinator): grep -n 'generated' engine/smogon_priors.js → :219; grep -n 'smogon-priors' engine/engine_release.js → :150 in SOURCES; the workflow → :17 cron 0 7 4 * *, :18 cron 0 7 11 * *, :83 git add data/smogon-stats data/smogon-priors.json data/smogon-priors-bo3.json data/mew.js; data/smogon-priors.json "generated":"2026-09-04"; git log -3 -- data/smogon-priors.json → 441196b8 2026-09-04, 89d86b93 2026-08-11, c1e2ec2f 2026-08-04, all abra-bot . The ingest workflow (ingest.yml:236-252 , per OPS) documents this exact failure for move-priors.json and fixed it by writing an .observed table; the smogon workflow was never given the same treatment.
- **What it could have put in front of a reader.** Three prior bot commits already moved this file; the next one lands mid-6.o.o and every release cut this week reads "1 of 27 files moved since".
- **Fixed? FIXED** (docs/_reports/2026-09-09-fix-smogon-cron.md). Workflow-only: the rebuild step now runs node engine/smogon_priors.js && cp -f data/smogon-priors.json data/smogon-priors.observed.json , then unconditionally git checkout -- data/smogon-priors.json , and the git add line stages the observed twin instead of the frozen source. The hazard was proven before the edit: a same-month run moved **2 lines** in each priors file, both the generated date, byte-identical with the date masked. The tree digest did not move — node engine/engine_release.js drift b730e44f3314 → **NO-DRIFT** (coordinator confirmed). tests/test-workflow-paths.js staging clause 4 of 4. data/smogon-priors.observed.json is seeded

and must be `git add`ed with the commit. `data/smogon-priors-bo3.json` stays on the direct path: it is NOT a frozen source (coordinator confirmed — `SOURCES` mentions it only in a comment; its readers are under `web/`).

- **Severity: BLOCKS 6.0.0** (time-boxed: Thursday 07:00 UTC). **Owner: OPS.**

6. 27 red tests, none waived by name, including a registered GATE at exit 1

- **What is wrong.** `cmd /c tools\lownode.cmd tests\run-all.js` (MEASURE §1, 16:03–16:48Z, log 848 lines, `EXIT=1`): **150 passed, 27 failed, 0 skipped** of 177. None is waived by Will by name in any record the review found. The session-close report carried three reds "openly"; the suite has 27.
- **Groups** (MEASURE §1): **A** — 8 files that are deliberate consequences of two owner decisions (web paused: `test-site-sync`, `test-site-data-fresh`, `test-web-quarantine-loaders`, `test-web-status`; MAG paused: `test-forced-switch`, `test-team-preview-race`, `test-wiring`, `engine/em_validation.js`) — these need a named waiver each, or the decisions reversed. **B** — 17 instrument / docs / store reds (`test-lownode` priority arm raced; `test-fixture-legality` walked another agent's worktree; `test-counter-init` two NaN counters; `test-mutation-coverage`; `test-prng`; `test-policy-promote`; `test-workflow-paths` "a store has rows in no shard"; `test-stadium-roster`; `test-docs-quarantine` — a withheld MILTANK figure in `docs/RUNNING-NOTES.md`:286; `test-quality` clean share 31.3% vs 28.0% recorded; `engine/selftest.js`; `engine/conformance.js`; `engine/provenance.js` red by construction until the re-run; `engine/validate_selfplay.js` 89 duplicate ids; `engine/sanity_check.py` 2 bad winners; `engine/identity_audit.js`; `test-board-browser`). **C** — 2 engine reds: `engine/gate_fail_and_silent.js` and `test-pinch-family` (1 of 61).
- **How the gate red was proved** (coordinator): `node engine/gate_fail_and_silent.js`; `echo $? → exit 1` (`LIVE 1` cause(s) over 1 game(s) — event missing from `medicham2` :: `|-fail|p2a <> |move|p2b|rockslide`). `node engine/where.js --gates` lists it under a CLOSED register row (#241) per the code review — a closed row whose regression gate is red.
- **What it could have put in front of a reader.** A major cut over 27 unwaived reds is "one of the known failures" twenty-seven times, which is the banned state by name.
- **Fixed?** Not by this session's three fixes, which touched one gate directly (`test-docs-current` 32/1 → 33/0) and one clause of `test-workflow-paths` (the store-currency clause, green after the recovery per the store report). The post-fix suite count is in §3.
- **Severity: BLOCKS 6.0.0** as a process fact. **Owners:** A — Will (waivers); B — MEASURE / OPS; C — ENGINE.

7. Five refuted causal claims stand in living documents, one in CLAUDE.md

- **What is wrong.** The causal audit classified 34 claims: 20 TESTED, 9 HYPOTHESIS, **5 REFUTED-BUT-STILL-STATED** (plus one in a live script header). Every refuted one is a claim about infrastructure — git, a timer, a heading — where the retraction registry cannot help because there is no number to strike.
- **The one the coordinator confirmed.** `sed -n '354,358p' CLAUDE.md → "The union driver is the confirmed cause of the store duplicating ... See CHANGELOG 3.1.2."` `sed -n '26611,26620p' CHANGELOG.md` (under 3.23.0) → "a fourth event — 009af26 'store: dedupe after rebase (137 duplicate lines)' — sits at depth 625, with no active union attribute, 208 commits after the driver was removed ... Recorded as a hypothesis with a known counter-example rather than as the cause. A fifth event is visible at depth 14: a commit published a store of 0 lines." CLAUDE.md states as confirmed what the CHANGELOG downgraded to hypothesis-with-counter-example. The forensics report goes further (§5 below): three of six duplications happened with no driver at all.
- **The other four** (per the causal report, file:line): `docs/ARCHITECTURE.md`:42-43 blames `merge -X ours`, the diagnosis 3.1.2 retracted the same day; `docs/SEARCH.md`:2506 says an auto-commit timer "publishes on a timer here" — dead since 2026-07-25, measured dead 2026-08-06; `docs/ENGINE.md`:404 heading "ABOUT ONE IN SIX" and :410-411, :419-420 "3 of the 19 causes ... 6.0.0 should say so in those words" — retracted to 2 of 19 at 5.273.0 and at :306-311 of the same file; `CLAUDE.md`:577-579 STALE marker says four MEDICHAM defects "were not re-checked" when the record shows all four re-checked (five living-document copies say Weather Ball, Sand Rush and Damp "are retracted as defects and are correct").
- **What it could have put in front of a reader.** The `ENGINE.md`:420 instruction is the dangerous copy: a 6.0.0 fold-in that follows it publishes a retracted rate.
- **Fixed? FIXED** (`docs/_reports/2026-09-09-fix-retractions.md` §A). All five rewritten across 8 locations: `CLAUDE.md` (union driver → hypothesis supported inside the driver's window and unsupported outside, the prohibition on restoring it kept verbatim, and the sharding cutover recorded as the sixth shape; the DAMP marker's "not re-checked" sentence replaced with

the four receipts), docs/ARCHITECTURE.md:42-43, docs/SEARCH.md:2506 (clause struck with the date), docs/ENGINE.md:404/412-413/418-419/423 (heading and instruction now read 2 of 19 per 5.273.0). tests/test-docs-current.js 33 passed / 0 failed after (coordinator re-ran). **One discrepancy with this review:** the record shows THREE doublings inside the union driver's window (7,040, 401 and 8,139 lines — CHANGELOG.md:26613-26615, .gitattributes:10-11), not two; CLAUDE.md now states the record. Not edited, reported: .gitattributes:9-11, ingest.yml:355-357 and push-all.bat:9-10 are live files, not living documents, and still name the driver as THE mechanism.

- **Severity: BLOCKS 6.o.o** for R4 (an instruction to publish a retracted number); **SHOULD FIX BEFORE** for the rest. **Owner: MEASURE** (documents), ENGINE (ENGINE.md).

8. The white paper states the opposite of three of its artifacts — thesis defence MAJOR REVISIONS

- **Coordinator confirmed all three.** (a) WAR: data/war.json:verdict = "WORSE THAN A COIN AT EVERY REGULARISATION STRENGTH TESTED ... lambda=200 at 0.69358 against a coin's 0.69315"; docs/ABRA-whitepaper.md:1968 quotes 0.7048, :1986 "WAR barely clears it". (b) NMF: engine/nmf_roles.py:154 ARCH_RANK = 6; data/nmf-roles.json generated 2026-08-04, archetype_rank 6, archetype_recon_error 0.682; data/nmf-rank-selection.json:shipped_rank stability 0.8148 against a null of 0.9218, excess_over_null -0.107; the white paper :1978 says " (error 0.53)" and :1981-1983 calls topic coherence "the next refinement". Two prior committees asked for this item. (c) CHOMP-EV: engine/chomp_ev.js:120 if (!CLEAN.has(g.id)) continue; sits ABOVE the qualify test at :128-129, so the selection audit can never receive a game; data/chomp-ev.json:selection_audit reads eval_mean_turns 8.12, excluded_mean_turns null; the white paper :1697-1699 reports "6.5 turns / 1280 rating vs 6.08 / 1267 excluded ... making the null conservative" — four numbers in no artifact, from a comparison never made.
- **What it could have put in front of a reader.** It already has. These are published sentences.
- **Fixed? FIXED as a retraction** (docs/_reports/2026-09-09-fix-retractions.md §B): 22 figures withdrawn or replaced across 21 locations in 7 documents, every replacement READ from the artifact on disk (war.json, nmf-roles.json, nmf-rank-selection.json, chomp-ev.json, policy-eval.json), none re-run. The derived retraction registry now refuses 35.9%, 71.6% and 1,205 repo-wide; 52.4% and 1280/1267 were deliberately NOT registered because a strong entry would accuse docs/MEASURE.md's majority-class cells and docs/ENGINE.md:35712 ("1,280 lines"). tests/test-docs-current.js 33/0. **The re-analyses remain owed** — T1-T5 in §8: the WAR split, the NMF rank, a real CHOMP-EV audit, the §1-ceiling artifact, and the store-layer re-run on quality filter 1.3.0 (every artifact read predates 2026-08-27).
- **Severity: BLOCKS 6.o.o** for the three sentences (a retraction never waits for a major; CLAUDE.md living-docs rule). The other nine defence items are **SHOULD FIX BEFORE**. **Owner: MEASURE**. See §9.

9. The store-duplication history: six duplications, the union driver supported for two, and no test asserts id uniqueness or monotone growth across a commit

- **What is wrong.** The forensics report walked all 101 commits of data/games.ladder.jsonl across all refs. Of six duplication events it counts on the ladder store, the union driver was active for three (#3, #4, #5) and a rebase replay is visible for two (#3, #5); three (#2, #6, #8) happened with no driver. The one merge commit in the whole history (1d70e0f6, blamed by CHANGELOG 2.10.0) added exactly 169 lines and duplicated nothing. No check anywhere asserts tracked-store $\supseteq$ previously-tracked-store, local plain store == HEAD shards, bo3 id uniqueness, or gz/shard integrity except when --verify-parsed is run by hand (report §1c). Per-incident table in §5.
- **What it could have put in front of a reader.** Finding 1 is the seventh row of this table. The bo3 store was ~40% duplicate lines for ~2.3 days (#7) and nothing said so.
- **Severity: SHOULD FIX BEFORE** (the check); the recovery is finding 1. **Owner: MEASURE**.

10. The engine work proper: 13 narration causes, the Infiltrator-through-screens hole, the substitute family

- **Narration count, reconciled.** status.js prints "12 of 961"; ENGINE reports "13 of 958 plus 3 void". Both are read from the same artifact and both are right about their population: mid_void.diverged_among_usable = 13 narration-only games among the 958 usable; engine/quarantine.js:2316 subtracts the one CLOSETED perish-drain row (a protocol declaration), so undeclared = 12; the clause's denominator is the artifact's games = 961, which includes the 3 void games. So "12 of 961" is a

958-population numerator over a 961-population denominator. Say "12 undeclared of 13 narration-only among 958 usable; 3 void" — never one bare ratio.

- **The 13 causes by mechanism** (ENGINE §1): the substitute family, 4 causes, one mechanism (this engine's arrival loop is INSIDE `_stepApply`, the authority's is outside `spreadMoveHit`; plan written in commit `b2508865`, not edited); the residual trio, 3 causes, UNEXPLAINED after six batches; Spicy Spray, 1, unblocked by reading `sim/battle.ts:789-790` (`DamagingHit` sorts by `compareLeftToRightOrder`, deterministic) with two false engine comments still standing at `:40323` and `:40250`; the perish `|upkeep| drain`, 1 (Perish Song 4,259 pool slots); Trick's `-fail`, 1 (a coarse mega-stone guard at `:31377` where the fine rule `itemRefusesTake` already exists at `:9909` — the FACTS-ARE-GLOBAL shape); post-KO switch-in order, 1; two singletons with derivations only.
- **Infiltrator through screens** (ENGINE §2): exactly one of 830 census rows carries `ignoresScreensAndSubs`, and it stages the substitute half. The screens half IS implemented (`:13785`) and has never been probed. A bypassed screen is a doubled damage roll — board-material. Four screens, not five: the fifth the brief named is `isNonstandard: "Past"` in the mod (`data/mods/champions/moves.ts:644-647`). The only tag whose name joins halves.
- **Severity**: the substitute family, Spicy Spray, perish drain, Trick, post-KO order — SHOULD FIX BEFORE; Infiltrator screens — BLOCKS 6.0.0 on the cheap-and-named grounds the session-close gave it; residual trio — DEFER WITH REASON (say the mutual-residual-KO corner is unmeasured). **Owner: ENGINE.**

11. The roster gate passes on 139 of 202 abilities compared

- **What is wrong.** `data/roster.abilities.json` (11:55:09Z, release `b730e44f3314`): matched 139, could-not-stage 44, control-not-quiet 14, deferred 5 (`scope: in_scope 202 = 139 + 5 + 14 + 44`). Items 142 of 148 (6 could-not-stage); moves 487 of 500 (10 could-not-stage, 3 deferred). The gate condition is "no FIRED-AND-BOARDS-DIFFER and no DID-NOT-FIRE"; COULD-NOT-STAGE and CONTROL-NOT-QUIET are neither. 32 of the 44 are "THE STAGING IS INERT" — a claim about the fixture, never about the mechanic. By pool usage the uncovered ones that matter: Compound Eyes (1,671 pool bodies, 0 census rows, roster inert), Imprison (1,046 slots) and Memento THREW on `pass, move 1` — a harness defect.
- **What it could have put in front of a reader.** A 6.0.0 sentence claiming 202 on evidence for 139.
- **Severity: BLOCKS 6.0.0** as wording (state the compared set), SHOULD FIX BEFORE for the Compound Eyes probe and the Imprison/Memento throw. **Owner: ENGINE.**

12. The Cowork / Claude Code boundary is convention, and there is a second publisher not in the table

- **Per the causal-and-boundary report:** of five boundary clauses, 1 is ENFORCED (push credentials), 1 is enforced for the race (git fast-forward + bound merge driver + serialised workflow), **4 are CONVENTION ONLY** (inbox-only, outbox-only, one-agent-at-a-time, suspect-figures). `git log -n 300 : 250` commits by Will Hooper, **50 by abra-bot**; the WHO MAY WRITE table (`CLAUDE.md:194-198`) does not list it, and the cadence it does state ("hourly") is stale — `ingest.yml:5-16` moved the cron to every six hours on 2026-08-21. The merge-driver binding is per-clone and manual; nothing tests that it is bound.
- **Two instances this session of the one-at-a-time clause failing silently:** a peer session was running on this repo when the review started (it landed `d6789951`); and a git worktree at `.claude/worktrees/agent-af59bcfe6a6444960` was walked by `tests/test-fixture-legality.js` as repo source, making the main tree's suite red (MEASURE §1.B, §12).
- **Severity: SHOULD FIX BEFORE** (table row, `orient.js` driver-binding clause, worktree exclusion in sweeps). **Owner: OPS / MEASURE.**

13. Documentation debt: 51 of 100 notes rows owed, 119 figure edits, 40 lift / 24 stay, and no basis declared

- **Coordinator:** `node engine/open_work.js` → 51 of 100 notes rows owed, documents last folded at 5.266.0, CHANGELOG top 5.274.0; `node engine/major_readiness.js` → living-doc rewrite 13 withheld + 84 stale + 22 untraceable = 119 figure edits across 5 documents; re-run 40 LIFT / 24 STAY (STAY = MAG family + MILTANK); VERDICT NOT READY.
- **MEASURE §6/§11:** clause 5d would REFUSE a 6.0.0 entry today (`major_without_basis`), while two rows already declaring Basis. CHANGED (`docs/RUNNING-NOTES.md:366`, `:444`) sit unversioned and would refuse any 5.x. 48 of 51 rows are [Unreleased] and unjudged. `engine/mew.js` is misfiled in LIFT — it loads MAG through `magnemite.js` (`:376`, `:661`) and MILTANK (`:550`); the predicate greps the generator's own source only.

- **Severity: BLOCKS 6.o.o** (write the basis row); SHOULD FIX BEFORE (mew.js to STAY; predicate follows one require hop). **Owner: MEASURE.**

14. The authority pin is stamped everywhere and consumed nowhere

- **Per the forensics and code reviews:** engine/champions_sim.js:85 pins 20ad99ffc9a5...; actualCommit() reads git rev-parse HEAD; **636 of 636** releases carry showdown_commit (grep -l '"showdown' data/releases/*/*.json | wc -l). commit_matches has **zero consumers** outside the file; engine_release.js:608 only console.error s on a re-cut against a different commit. The checkout is not locally modified (git -C ../pokemon-showdown status --short → empty) and is 72 commits behind upstream as of the 2026-08-31 fetch. **4,335** <file>.ts:<line> citations (engine 1,266; tests 965; docs 2,104) are valid exactly as long as the checkout stays put, and none is content-verified.
- **Severity: SHOULD FIX BEFORE** (consume commit_matches in the differential's preflight; a --fail-on-drift on cut). **Owner: ENGINE.**

15. Everything else the reports carry, in one pass

item	receipt	severity	owner
The test suite and instruments write to the tree: 15 tracked files modified by the run, including data/mechanics-census.json rewritten by tests/test-arm-steering.js (11:54:09Z → 16:21:25Z) — a pinned gate input	MEASURE §12	SHOULD FIX BEFORE	MEASURE
The gate run carried no census pin: steering.pinned false (coordinator read)	data/game-differential.json:steering.pinned	SHOULD FIX BEFORE (pin on the re-run)	MEASURE
A frozen release still reads the live stores: set_priors.js:286-296 observedSets() opens data/games.bo3.jsonl / games.ots.jsonl at play time; --team-store pins the POOL, not the SET FILLER	code review §1.6	SHOULD FIX BEFORE	ENGINE
engine/mew.js and test-wiring.js need a 6 GB heap and declare none; test-wiring red on the real tree (10 NOT WIRED, "consistent with, not proven" the MAG weights refusal)	test-breaks §3.1, MEASURE §1.A	SHOULD FIX BEFORE (re-run the day the weights load)	MEASURE
The oldest release d3d04b669e18 fails its own manifest in the worktree (medicham2-browser.js manifest 645b93a3878f, snapshot f4bfd9b997d); real tree unverified	test-breaks §3.3	SHOULD FIX BEFORE (verify)	MEASURE
Two hand-typed Pokémon tables in code: MEGA_ABIL (medicham2-browser.js:8191-8207, has a comparator in test-effective-identity.js); the 18×18 type chart TC in engine/validate_damage.js:10-25 inside a GATE, no comparator	code review §3	SHOULD FIX BEFORE (derive)	ENGINE
45 of 237 MEDI_* knobs are named in no test — their red arm exists only in prose	code review §2	SHOULD FIX BEFORE	ENGINE
17 tests/probe_*.js are run by nothing (not the suite, not the register); data/mutation-coverage.json last generated 2026-08-22	code review §2	SHOULD FIX BEFORE	MEASURE
Three readers, three meanings for exit codes: run-all (2 = SKIP), register_reality.js (2 = UNDECLARED unless declared), wire_ladder.js (4 = refused) — ROADMAP #380	code review §1.1	SHOULD FIX BEFORE	MEASURE
engine/where.js --artifacts WHO-WRITES index is a 400-char proximity heuristic with verified false positives	code review §4	DEFER WITH REASON	MEASURE
The pinned pool is gitignored — one laptop copy; only FROZEN.md digests are tracked	OPS §3	SHOULD FIX BEFORE (copy out, verify sha256)	OPS

item	receipt	severity	owner
Local plain stores are five days behind origin (last written 2026-09-04 01:30); every unpinned local reader is behind	OPS §3, forensics §1d	SHOULD FIX BEFORE (-- restore-parsed)	OPS
The legality verdict is a 2026-08-27 snapshot over 67,384 games; games since are unjudged	OPS §8	SHOULD FIX BEFORE (refresh before any clean-store figure)	OPS
The live bot loads the quarantined weights; nothing shows it running since 2026-08-04	OPS §5	SHOULD FIX BEFORE (say so in the release)	OPS
Self-play store: 89 duplicate ids (#536); the generator is fixed, the store is not	OPS §4, forensics §1d	DEFER WITH REASON (quarantined, nothing in 6.0.0 reads it)	OPS
register-reality.json verdicts (2026-09-08T16:34Z) are a day and seven engine commits older than the engine they clear	MEASURE §12	SHOULD FIX BEFORE	MEASURE
docs/THESIS-DEFENCE-REVIEW-2026-09-09.md has no version header and is not in the docs-currency baseline — clause 2b red on the next suite run	MEASURE §11	SHOULD FIX BEFORE	MEASURE
Seven of eighteen named register rows are closed on the tree and open in the register (#315 refuted for Champions, #362, #393, #403, #319, #318, #528)	ENGINE §6	DEFER WITH REASON (hygiene)	ENGINE
Two empty catch(e){} around tag reads (:5500 , :6275): priority blocking silently becomes an empty map	ENGINE §10	SHOULD FIX BEFORE	ENGINE
tools/lownode.cmd is LF on disk; LESSONS records an LF .cmd opening a prompt instead of running; test-lownode has no arm for the :derive path	code review §7	SHOULD FIX BEFORE	OPS
380 split('\n') sites vs 20 split(/\r?\n/); 14 files pair a file read with a \$-anchored regex; the docs_scan blindness was this class	code review §7	DEFER WITH REASON (one readText())	MEASURE
Damage differential has never applied a multi-hit move (skipped_multihit 134 , skipped_ability_multihit 17); covered by roster + probe + census; declared band gap	ENGINE §5	DEFER WITH REASON (state it)	ENGINE
The 100 MB wall is retired for the store: the .gz is untracked; largest tracked blob data/games.r4-decided.jsonl 44,940,683 B; pack still grows ~1.2 MB/day	MEASURE §12, forensics §1e	none	OPS

Changes made

Every file touched this session, what changed and why, from `git diff --stat` and `git status --short` read at fill time (15 tracked files modified, 192 insertions / 87 deletions; 21 untracked additions) plus the three fix reports. Nothing is committed at fill time. Test results before, from MEASURE §1: **150 passed / 27 failed / 0 skipped of 177**, EXIT=1 . After:

151 passed / 26 failed / 0 skipped of 177, EXIT=1 (`cmd /c tools\lownode.cmd tests\run-all.js` from PowerShell, 838 lines of output, after every fix above and after the restamp). Three reds went green: `tests/test-lownode.js` , `tests/test-policy-promote.js` , `tests/test-workflow-paths.js` (the last because the shard-superset guard is what it wanted). Two went red, **both caused by this session's own store recovery**: `tests/test-medicham-coverage.js` and `tests/test-degradation-budgets.js` died at the default Node heap (exit 134) because `data/games.bo3.jsonl` grew to 32,801 rows / 298 MB and both read a store whole. Four scripts already red for their own reasons (`tests/test-quality.js` , `engine/selftest.js` , `engine/provenance.js` , `tests/test-site-data-fresh.js` 's

child) also began dying of heap instead of reaching their verdict. All six now carry an ABRA-HEAP: 4096 declaration that `tools/lownode.cmd` and `tests/run-all.js` honour, `engine/status.js` passes the child's declaration when it spawns `engine/provenance.js`, and each was re-run individually: `medicham-coverage` 134 → 0, `degradation-budgets` 134 → 0 (11 of 11), `provenance` 134 → 0, `quality` 134 → 1 (its own pre-existing red), `selftest` 134 → 1 (its own), `site-data-fresh` 1 → 1 with its "artifacts current" clause no longer passing vacuously on an empty provenance text (`docs/_reports/2026-09-09-fix-heap-declarations.md`). The suite was NOT run a third time; the after figure stands at 151 / 26 with those six re-run singly. The remaining 26 reds are listed by name in `docs/_reports/2026-09-09-pre-600-measure-review.md` §1 and none is waived; the worse half of the provenance finding — a plain `node engine/status.js --write` restamping every ledger with `provenance: NOT DERIVED` and exiting 0 — is closed by the spawn fix.

Version. MINOR 5.275.0 — 22 published figures moved under an unchanged basis (SemVer clause 7; the notes rows declare Basis. unchanged). No engine byte moved: `node engine/engine_release.js drift b730e44f3314` → NO-DRIFT (coordinator confirmed), so no release re-cut is owed and every current gate artifact still stamps a release that matches the tree.

3.1 The store recovery (finding 1) — `docs/_reports/2026-09-09-fix-store-recovery.md`

file	diff	what changed and why
<code>data/parsed/games.ladder/20260909T1708-00.jsonl.gz</code>	new, untracked, 5,287 rows	recovered ladder rows, date span 2026-08-22 → 09-01; three ladder shards rather than one because the 32 MiB source cap split 66.5 MB of source
<code>data/parsed/games.ladder/20260909T1708-01.jsonl.gz</code>	new, untracked, 5,414 rows	same, 2026-09-01 → 09-06
<code>data/parsed/games.ladder/20260909T1708-02.jsonl.gz</code>	new, untracked, 409 rows	same, 2026-09-05 → 09-06; the three carry +11,110 rows in total
<code>data/parsed/games.bo3/20260909T1708-00.jsonl.gz</code>	new, untracked, 3,457 rows	recovered bo3 rows, 2026-08-22 → 09-04
<code>data/parsed/games.bo3/20260909T1708-01.jsonl.gz</code>	new, untracked, 1,295 rows	same, 2026-09-04 → 09-05; the two carry +4,752 rows in total
<code>build/compress-stores.js</code>	63 (+45 / -18)	--check compared the shards to the LOCAL plain file, which was the thing that was wrong. It now also reads the ids git tracked at HEAD~1 — both forms, <code>data/parsed/<store>/*.jsonl.gz</code> and the retired <code>data/<store>.jsonl.gz</code> , so a cutover between forms is inside the claim — and fails if any is absent from the shards on disk; it THROWS when HEAD~1 cannot be read rather than passing quietly. The 18 deletions are <code>shardedCount()</code> and its comment, which existed for a 58 s measurement that re-measures at 2.2 s today. No new flag, file or knob.
<code>engine/dedupe_store.py</code>	5 (+3 / -2)	wrote CRLF on Windows (<code>os.fdupen(fd, 'w', encoding='utf-8')</code> → <code>os.linesep</code>); shown red on a 5-row probe (5 CR bytes, sha moved) and fixed with <code>newline='\n'</code> BEFORE it touched the real store. Unfixed, the 15,862 recovered rows would have been published CRLF and --verify-parsed would have failed.

Method: --restore-parsed the local plain stores to HEAD's shards (76,833 → 81,269 ladder, 25,522 → 28,049 bo3; the local file was a strict subset), append the `d2a418a5` monoliths, dedupe keeping the first occurrence so every shared id keeps its tracked shard bytes (169,615 lines → 92,379 unique ladder; 59,335 → 32,801 bo3; 0 unparseable), cut shards. ots needed nothing: its shard and the monolith decompress to the same sha256 (`cd21077a4578afa3`).

Receipts, before → after: tracked ids ladder **81,269** → **92,379**, bo3 **28,049** → **32,801**, ots 4,167 unchanged (coordinator confirmed). `comm -23` of `d2a418a5` 's ids against all shards: **0 / 0 / 0** (coordinator re-ran ladder and bo3). --verify-parsed IDENTICAL on all three stores (ladder `cde0fa05d5177469`, 25 shards, 481,072,929 B; bo3 `072c4c61ed23bea3`, 19 shards, 298,313,305 B). `git diff --stat HEAD -- data/parsed` empty — no existing shard changed (coordinator confirmed). The new --check clause: green on the tree in 8.7 s (carries every id HEAD~1 tracked (81269 / 4167 / 28049)); **red** on a one-row sandbox break (LOST 1

id(s) , exit 1); **red** on a worktree replay of 18432bcb with HEAD~1 = 0601a600 (LOST 11513 ladder, LOST 5764 bo3, exit 1 — the loss AT the cutover; later ingests re-pulled 403 and 1,012 from the rolling replay pool, leaving today's 11,110 / 4,752). tests/test-workflow-paths.js 5 passed / 0 failed, 9.3 s. engine/sanity_check.py 95/1 → 94/2: the added red is brought = 6 on both sides of gen9championsvgc2026regmb-2676161109 , a record d2a418a5 already tracked — the recovery surfaced a pre-existing ingest defect, it did not create one.

Still owed (store report §11): wire --check into ingest.yml after its shard step (on a fetch-depth: 1 checkout it throws on HEAD~1 , which is the loud answer); the runner's shrink guard still measures against itself; sanity_check.py stays red on 2 parser clauses (OPS).

3.2 The Smogon cron (finding 5) — docs/_reports/2026-09-09-fix-smogon-cron.md

file	diff	what changed and why
.github/workflows/smogon-stats.yml	19 (+16 / -3)	the rebuild step now runs node engine/smogon_priors.js && cp -f data/smogon-priors.json data/smogon-priors.observed.json followed by an unconditional git checkout -- data/smogon-priors.json ; the git add line stages the observed twin instead of the frozen source; a comment block states the defect, the proof and where promotion lives (a hand run of node engine/smogon_priors.js , the command that already exists). Done in the workflow and not the script because engine/smogon_priors.js is itself in SOURCES (entry :138) and a flag would have moved the digest and stranded b730e44f3314 — moving the digest to stop the digest moving is the wrong trade.
data/smogon-priors.observed.json	new, untracked, 1.3 MB	seeded from the same run (generated: 2026-09-09) so the path exists on the first cron run even if the derivation fails — a git add on a missing path exits 128 under bash -e , the exact test-workflow-paths defect. Must be added with the commit.

Receipts: the hazard proven before the edit — a same-month run (2026-08 in, 2026-08 out) moved **2 lines** in data/smogon-priors.json and 2 in -bo3.json , both the generated date, JSON byte-identical with the date masked; the three bot commits (08-04, 08-11, 09-04) were each that rewrite. After the edit: node engine/engine_release.js drift b730e44f3314 → **NO-DRIFT** (coordinator confirmed). YAML parses, six steps. tests/test-workflow-paths.js staging clause **4 of 4**, including smogon-stats.yml stages data/smogon-priors.observed.json . data/smogon-priors-bo3.json stays on the direct path — it is not a frozen source (coordinator confirmed: SOURCES names it only in a comment; its readers are under web/). Nothing under engine/ edited.

3.3 The retractions (findings 7 and 8) — docs/_reports/2026-09-09-fix-retractions.md

file	diff	what changed and why
CLAUDE.md	23 (+20 / -3)	R1: the union driver "confirmed cause" → hypothesis supported inside the driver's window (three doublings, not two — see finding 7) and unsupported outside, prohibition on restoring the driver kept verbatim, the sharding cutover added as the sixth shape. R5: the DAMP marker's "not re-checked" sentence replaced with the four receipts (3.79.0, ROADMAP #100, chargeSkippedByWeather , tests/probe_transform_faint_revert.js); the dated 2026-08-08 paragraph untouched.
docs/ARCHITECTURE.md	8	R2: "7,040 duplicates from merge -X ours " → divergent reconciliations, with 3.1.2's withdrawal and 3.23.0's counter-example in a parenthetical; header v1.2 pin untouched
docs/SEARCH.md	4	R3: the auto-commit-timer clause struck (~~~~) with the date; the digest-set conclusion stands on the dirty-tree reason alone
docs/ENGINE.md	17	R4: "ABOUT ONE IN SIX", "3 of the 19 causes", the Instruct/perish survivor and "about 3 causes ... 6.0.0 should say so" each struck and restated as 2 of 19 with the 5.273.0 receipt; the instruction to 6.0.0 now carries the right number
docs/ABRA-whitepaper.md	70 (+43 / -27)	6 figure locations: WAR (:1968 , :1986), NMF (:1978 , :1981-1983), CHOMP-EV (:1691-1698), XATU (:1676-1678), §1 ceiling (:1491-1495) — see §4
docs/MODELS.md	9	4 locations: WAR row struck and artifact row added (:1783), NMF (:1798), CHOMP-EV (:1687), XATU (:1405)
docs/ROLE-FAMILY.md	20	4 locations: WAR (:62-63 , :98-99), NMF (:81 , :102-104)

file	diff	what changed and why
docs/PUBLICATION.md	13	3 locations: WAR item heading (:31-32), CHOMP-EV n (:21), §1 ceiling (:16)
docs/SUMMARY.md	7	2 locations: WAR (:1310), §1 ceiling 52.4% (:1392)
docs/DEFENSE.md	3	1 location: :247 35.9% → 29.8% with the artifact field (a living document, corrected in place when the registry accused it)
docs/MEASURE.md	5	1 dated correction note at :5337 , where a 71.6% is PORY-NN's auc — a different measurement — so the registry does not accuse it
data/docs-currency-baseline.json	13	unversioned_exempt gains the two 2026-09-09 reviews; known.retraction_violations gains three entries for the dated thesis review (1,205 , 35.9% , 71.6%), on the 07-28 review's precedent; the test itself rewrote generated → 2026-09-09T17:12:42Z , changelog_top_at_baseline 5.270.0 → 5.274.0, and dropped two citation_mismatches (the XATU paragraph is now a QUALIFIED block) — its ratchet behaviour on a green run

Receipts: **5 causal claims** rewritten across 8 locations; **22 numeric values** withdrawn or replaced across **21 figure locations** in 7 documents (whitepaper 6, MODELS 4, ROLE-FAMILY 4, PUBLICATION 3, SUMMARY 2, DEFENSE 1, MEASURE 1 note). Every living document in the brief's list was grepped for every old value; the deck, technical docs, README, SLOWKING paper and the OPS / WEB ledgers carried none. `node tests/test-docs-current.js` **32 passed / 1 failed → 33 passed / 0 failed** (coordinator re-ran; the 1 was clause 2b, the undeclared 09-09 review, closed by the baseline entry). An intermediate run went red on 3b(a) with 4 NEW restatements produced by the registration itself — `DEFENSE.md:247` and three lines of the dated thesis review — and was closed in the same pass; none is left standing and none was waived. `node engine/docs_scan.js --quarantine:23` untraceable across 3 documents before and after — no living document gained an untraceable figure. `--owed:51` of 100, unchanged (no row written by this pass). Every replacement was READ from an artifact already on disk; nothing re-ran a generator.

3.4 Review deliverables and reports (new, untracked)

file	source
docs/ARCHITECTURE-REVIEW-2026-09-09.md	this file (no PDF on disk at fill time)
docs/THESIS-DEFENCE-REVIEW-2026-09-09.md , .pdf	the thesis defence agent; PDF via headless Chrome (§6). No .html is on disk, contrary to the draft's earlier {md,html,pdf} .
docs/_reports/2026-09-09-pre-600-{engine-review,measure-review,ops-review,code-review,test-breaks,store-and-authority,causal-and-boundary}.md (7)	the read-only division agents
docs/_reports/2026-09-09-thesis-defence-notes.md	the thesis defence agent
docs/_reports/2026-09-09-fix-{store-recovery,smogon-cron,retractions}.md (3)	the three fix agents

Twelve `_reports` files in all (the record-keeping pass added `2026-09-09-fix-rows.md`). The 15 tracked `data/*` files MEASURE §12 reported rewritten by the suite run were reverted by the coordinator with `git checkout -- data/` before the fix pass began (they are the suite's incidental rewrites of `data/mechanics-census.json`, `data/provenance-stamp.json`, `data/open-work.json` and twelve more, not an outcome); the final suite run at the end of the session was reverted the same way, so no suite-written artifact is committed with this release.

3.5 Landed by the record-keeping pass this session

`docs/RUNNING-NOTES.md` (three rows: store recovery, cron, retractions — all `Basis.` unchanged), `CHANGELOG.md` (the `[5.275.0]` entry), `docs/ROADMAP.md` (the store-loss DEFECT row, #550 or the next free id), and the division ledgers `docs/{ENGINE,MEASURE,OPS}.md`. Those files are owned by a concurrent rows agent and are not in the `git diff --stat` read above.

Numbers corrected

The table below is the NUMBERS CORRECTED table of docs/_reports/2026-09-09-fix-retractions.md §B, verbatim. Every artifact field was opened with `node -e` in that pass (commands at the report's §E). Every living document in the brief's list was grepped for each old value — white paper, deck, technical docs, SUMMARY, MODELS, ROLE-FAMILY, SLOWKING-whitepaper, PUBLICATION, README, ENGINE, MEASURE, SEARCH, OPS, WEB — plus CLAUDE.md, ARCHITECTURE.md and DEFENSE.md. "Checked-and-clean" lists the documents that were searched and carried NO copy.

old value	new value, or WITHDRAWN (reason; artifact field)	every file:line it appeared in (all edited)	checked-and-clean
WAR clean-store 0.7048 log-loss, accuracy 0.502 (the v3.2.0 run, 1,061 games)	0.6936 vs coin 0.6931, accuracy 0.504 , n_games 3,663 , $\lambda=200$ selected on held-out — <code>data/war.json:held_out.log_loss / .coin / .accuracy, n_games, ridge; verdict "WORSE THAN A COIN AT EVERY REGULARISATION STRENGTH TESTED"</code>	whitepaper:1968; MODELS.md:1783 (table row struck, artifact row added); SUMMARY.md:1310; ROLE-FAMILY.md:62; PUBLICATION.md:32	deck, technical-docs, README, SLOWKING-whitepaper, ENGINE, MEASURE, SEARCH, OPS, WEB. (MODELS.md:1775 restates 0.6905 only inside an explicit "The previous entry read:" quotation — left.)
"which specific species you bring at preview carries a small real signal "	WITHDRAWN — <code>war.json:verdict</code> : "Do not quote WAR as evidence that a species wins games"	whitepaper:1968; ROLE-FAMILY.md:63; PUBLICATION.md:31 (item heading)	deck, technical-docs, README, SUMMARY (already "Null"), SLOWKING-whitepaper, ledgers
"WAR barely clears it" / "WAR only edges it" / "beats the rating baseline (0.6905) "	WITHDRAWN — <code>war.json</code> contains no rating baseline; verdict as above	whitepaper:1986; ROLE-FAMILY.md:98-99; ROLE-FAMILY.md:62-63	all others
NMF archetype reconstruction error 0.53	0.682 — <code>data/nmf-roles.json:archetype_recon_error</code> (generated 2026-08-04, <code>archetype_rank 6</code>)	whitepaper:1978; MODELS.md:1798; ROLE-FAMILY.md:81	SUMMARY.md:1311 and PUBLICATION.md:41 already record 0.53 as withdrawn; deck, technical-docs, README, ledgers
"topic coherence (Mimno 2011) ... the next refinement"	DELETED; replaced by the criterion actually run: rank 6 excess over null -0.107 (stability 0.8148 vs null 0.9218), rank 4 most reproducible +0.0775 — <code>data/nmf-rank-selection.json:shipped_rank, most_reproducible, results</code>	whitepaper:1981-1983; MODELS.md:1798; ROLE-FAMILY.md:102-104	all others
CHOMP-EV "On 1,205 games"	1,102 — <code>data/chomp-ev.json:n_test</code>	whitepaper:1691; MODELS.md:1687; PUBLICATION.md:21	deck, technical-docs, SUMMARY, README, ledgers
CHOMP-EV log-loss 0.6918	0.6921 — <code>chomp-ev.json:proper_score_logloss.chomp_align</code> (coin 0.6931 unchanged)	whitepaper:1692; MODELS.md:1687	all others
CHOMP-EV sign test 0.512, CI [0.493, 0.535]	0.5123 [0.4997, 0.5246] — <code>chomp-ev.json:headline_beat_test.p_winner_more_aligned / .ci95</code>	whitepaper:1694; MODELS.md:1687	all others
CHOMP-EV forfeit-robustness 0.505	0.5082 — <code>chomp-ev.json:robustness_no_forfeits.p_winner_more_aligned</code>	whitepaper:1694	MODELS.md:1687 states it without a number (kept, number added)
CHOMP-EV selection audit "eval 6.5 turns /	WITHDRAWN — <code>chomp-ev.json:selection_audit.n_excluded_human = 0, excluded_mean_turns = null: engine/chomp_ev.js:120 if</code>	whitepaper:1696-1698; MODELS.md:1687 ("null is conservative")	PUBLICATION.md:21-22 carries no audit claim; deck,

old value	new value, or WITHDRAWN (reason; artifact field)	every file:line it appeared in (all edited)	checked-and-clean
1280 rating vs 6.08 / 1267 excluded ... making the null conservative"	(!CLEAN.has(g.id)) continue; sits above the qualify test at :128 , so audExcl can never receive a game. No artifact establishes the bias direction. The thesis-notes measurement (7.81 vs 6.74 turns, +4.44 points) is referenced by PATH only, not restated — see the report's \$E for why.		technical-docs, SUMMARY, README, ledgers
XATU clone top-1 35.9% (CI 35.2–36.5), top-3 71.6% , CE 2.27 , baselines 4.54 / 2.91	WITHDRAWN — in no artifact. data/policy-eval.json:species_only_clone (committed 2026-07-31) reads top-1 0.2979 [0.2914, 0.3045] , top-3 0.6564 , CE 2.6353 ; baselines.global_move_freq_ce 4.7346 , uniform_moveset_ce 3.0286 — stated beside the withdrawal	whitepaper:1676-1678; MODELS.md:1405; DEFENSE.md:247 (35.9% → 29.8% with the artifact field)	SUMMARY (no copy today, despite MEASURE.md:5337 saying it quotes one), deck, technical-docs, README. MEASURE.md:5337's 71.6% is PORY-NN's auc, a different measurement (already recorded at MEASURE.md:5239); a dated correction note was added there so the registry does not accuse it.
\$1 ceiling Elo log-loss 0.687 vs coin 0.693 on 600+ games	WITHDRAWN — engine/predictability.py has no json.dump (grep: none); data/skill-variance.json and engine/skill_variance.py do not exist (ls)	whitepaper:1491-1492; PUBLICATION.md:16	MODELS.md:1404 "player-Elo ≈ coin (0.687)" sits inside a paragraph headed "The 2026-07-23 reading, superseded , kept" — a dated record, left. deck, technical-docs, SUMMARY, README, ledgers clean.
\$1 ceiling 52.4% , 95% CI [49.9, 54.9]	WITHDRAWN — same reason	whitepaper:1494-1495; SUMMARY.md:1392	PUBLICATION.md has no copy. MEASURE.md:6271, :6272, :6342 write "52.4%" as the majority-class baseline of the turn-o leaf arms — a different quantity — left, and deliberately NOT registered.

Counts: **5 causal claims** rewritten across 8 locations; **22 numeric values** withdrawn or replaced across **21 figure locations** in 7 documents. The derived retraction registry (engine/docs_scan.js:585) now refuses **35.9%, 71.6% and 1,205** repo-wide, from the living documents' own struck lines. **52.4%** and **1280 / 1267** were deliberately NOT registered: a strong entry would accuse docs/MEASURE.md:6271-6272, 6342 (the turn-o arms' majority-class baseline) and docs/ENGINE.md:35712 ("1,280 lines into an unrelated file") — the 9.7%-accuses-10% failure. Both are named as literals only in the CHANGELOG (exempt); the report proposes phrase-scoped entries for the hand-typed **RETRACTED** list in tests/test-docs-current.js:103 , not written (no test edits in that brief).

Identified by the reviews and NOT corrected this pass (out of scope for the retraction brief; listed in the report's OWED block and in §8):

figure	document says	artifact says	where
GURU predictive log-loss	0.7122	predictive_test.log_loss_matchup_prior 0.7124	whitepaper :1667-1672
SLOWKING archetype count / uniform	13 / 0.109 / TrickRoom→HyperOffense→Sand / 51% n=236	12 / 0.0761 / supported: false / 0.538 [0.466, 0.609] n=182	whitepaper :1716-1729

figure	document says	artifact says	where
exploitability / cycle / Rain vs Sun			
role- matchup median cell	n = 20 across 1,051 cells	1,101 cells, median n = 74 (computed by the defence)	whitepaper :1954 ; docs/SUMMARY.md:1309
functional roles	26	52 in ROLE_SIGNALS ; 46 matrix columns	whitepaper :1945 ; ROLE-FAMILY.md:25 ; SUMMARY.md:1309
ROLE- FAMILY §4 archetype table and names	the 2026-07-24 run's	data/nmf-roles.json:archetypes	docs/ROLE-FAMILY.md §4; whitepaper, MODELS
store sharding	"Supersedes. Nothing."	11,110 + 4,752 ids dropped from the tracked store — superseded by the record-keeping pass's store-recovery row (§3.5)	docs/RUNNING- NOTES.md:573 block
MILTANK profile figures in a notes row	66.6%, 27.1%, 92.0%, 25514, 42510, 99.5%, 78.8%, 83.7%	withheld (STAY artifact data/search-decision-profile.json)	docs/RUNNING- NOTES.md:286
SOURCES count	"twenty-five"	27 (require(' ./engine/engine_release.js ').SOURCES.length)	CLAUDE.md ; .gitattributes says 26
ingest cadence	"hourly"	every six hours since 2026-08-21 (ingest.yml:5-16)	CLAUDE.md:286 , :706

Causal claims audit

Every "X caused Y" the causal report classified, and the store-duplication history built from the forensics report's per-incident table — never from a handoff note.

Classification (causal-and-boundary report, 34 claims read in context of ~90; 2,186 causal-phrase lines seen, 295 strong-form; docs/ENGINE.md 's 167 strong-form lines SAMPLED, not audited)

class	count	examples with receipt
DISPROVEN, still stated	5 (+1 script header)	R1 union driver "confirmed cause" (CLAUDE.md:356-357 vs CHANGELOG 3.23.0 — coordinator confirmed); R2 merge -X ours (ARCHITECTURE.md:42-43 vs 3.1.2's same-day retraction); R3 the auto-commit timer (SEARCH.md:2506 vs CLAUDE.md:330-335); R4 "one in six" (ENGINE.md:404,410-411,419-420 vs 5.273.0 and ENGINE.md:306-311); R5 "not re-checked" STALE marker (CLAUDE.md:577-579 vs five living-doc copies); R6 push-all.bat:9-10
UNVERIFIED (hypothesis stated as fact)	9	H1 the driver/-X ours pair (5 copies); H2 DODUO "the constraint is the objective" (quarantined anyway); H3 concurrent runs "pushed RAM toward swap" (CLAUDE.md:167-171 , no RAM measurement cited); H4 statistical models "never touch damage, which is why" (ARCHITECTURE.md:229); H5 open sheets ~1% "need both players to agree" (technical-docs:1697); H6 "six of those thirteen games have a second thing wrong" (deck:213 , no field cited); H7 Sand Rush 140/280 reading left open (ENGINE.md:26795-26796); H8 dead API vs quiet day (OPS.md:87); H9 "hourly" ingest (CLAUDE.md:286)
VERIFIED	20	speed ties are the SORT (knob MEDI_ENTRY_STABLE_SORT=1 , 4 copies); 777 vs 961 is the --games flag (reproduced, 5.79.0); blank gate clauses were a line ending (stripCR , pinned both directions); joint 110 / empirical 35 attributed by knob-cleared runs on one release; brought inflated by forme changes (1,003 of 1,033 offenders contain mega); the speed-ratio series re-measured over 8 contention-free reps; the event die "translating rather than re-drawing" (closed-form)

Pattern the report states once and this review endorses: every DISPROVEN item is about infrastructure. The retraction registry (`engine/docs_scan.js:585` , `:689`) works for FIGURES and cannot see a MECHANISM, which is why R1–R3 survived — there is no number in them to strike.

The store duplications, per incident (forensics report §1a; `git log --all --full-history` , `git cat-file -s` , `awk` over blobs, `git show <h>:.gitattributes` for the driver)

#	when	commit	lines/ids parent → this	driver active	verdict on "the union driver did it"
1	07-24 03:31	b66aaecb	6539/6539 → 6645/6639 (+6 dup)	no (12 h before it existed)	UNSUPPORTED; not in the CHANGELOG
2	07-24 15:01	e59c3bda	6645/6639 → 7449/7042 (+401 dup)	no — added two minutes later	UNSUPPORTED; single-parent, author = committer
3	07-24 15:43	072e00d2 , 34532938	7449/7042 → 14188/7142 (+7046 dup); bo3 1492/897	yes	SUPPORTED for union-on-rebase (replays). The merge <code>1d70e0f6</code> that 2.10.0 blamed added exactly 169 lines and duplicated nothing
4	07-24 18:04	71de227c	7716/7716 → 7948/7547 (+401 dup, −169 ids)	yes	PARTIAL; no reconciliation commit; the store LOST 169 games; mechanism not recoverable
5	07-24 20:12	8917698d , 608288de	→ 16540/8000 and 16139/8000	yes	SUPPORTED; 3.1.2's "16,139 → 8,000"; the 45-commit rebase stopped at 43 is real
6	07-24 21:07	ceaf5c1b	8356/8356 → 8757/8356 (+401 dup)	no (removed 35 min earlier)	UNSUPPORTED; 3.3.0's "8,757 unique, 0 duplicates ... removal is holding" is true of a DIFFERENT store with the same number
7	07-24 → 07-26	bo3 store	595 duplicate ids from #3 persisted through every ladder dedupe until <code>72066189</code>	—	consequence of #3; <code>dedupe_store.py</code> hardcoded to the ladder path; bo3 ~40% duplicate for ~2.3 days
8	07-26 12:42	6d1a3627	14447/14447 → 14931/14794 (+137 dup)	no	UNSUPPORTED; 3.23.0's counter-example, confirmed
9	07-29 → 08-21	(no commits)	—	—	not a duplication: a 24-day hole in the tracked store, closed by <code>6ecc96de</code> "the only complete copy was on one laptop"
10	09-06 16:25	18432bcb	88,346 ids → 76,833 ids (shards from the local file)	—	NEW, UNRECORDED, OPEN — finding 1

Aggregate. The report counts **six** duplication events on the ladder store (#2, #3, #4, #5, #6,

8); #1 (+6 duplicate lines) is a seventh row with duplicate lines gained that the report's aggregate

does not count — flagged here, not resolved. Driver active for three (#3, #4, #5); a rebase replay visible for two (#3, #5); three with no driver (#2, #6, #8). So `.gitattributes` ' sentence "Duplicates have only ever entered this repository through git reconciliation" is not demonstrable from the repository; 2.10.0's `merge -X ours` diagnosis is UNSUPPORTED; 3.1.2's re-diagnosis is SUPPORTED for #3 and #5 and over-general for the rest. **Not reproduced:** 3.23.0's "a fifth event at depth 14: a commit published a store of 0 lines" — no zero-byte blob exists for any of the four store paths across all refs, and the depth-14 ancestor holds 70,278,317 / 17,138,942 / 31,923,870 bytes. **Removing the driver was correct. Calling it THE cause is not supported.**

What I could not do, and why

Honest and complete. Each line is an instrument that did not run, a claim that could not be confirmed, or a disagreement that was not resolved.

Instruments that died or were not run

- `node engine/engine_release.js census` (dry run) died at `FATAL ERROR: Reached heap limit` after 177 s at the 2 GB default; `node --max-old-space-size=3072 engine/engine_release.js compat engine/medicham2-browser.js` died the same way. Neither declares `ABRA-HEAP`. **The stranded-release count across all 636 releases is NOT DERIVED**; the last `data/release-census.json` is dated 2026-08-11 (118 releases). `tests/test-artifact-rerunnable.js` (heap 6144) passed inside the suite — 99 stamped artifacts over 36 releases, 57 re-runnable, 1 STRANDED and undeclared (known, was 1) — so the ratchet that matters holds (MEASURE §4).
- No engine script that plays a game was run by this draft, `status.js` was not re-run, and the differential was not re-run. The gate reading is the coordinator's 11:59 print and MEASURE's `medichamIsCorrect()` call at 16:35Z.
- None of the thesis's store-derived generators was re-run (seven agents live; each rewrites its artifact unconditionally; ROADMAP #288 — no store digest — means a re-run would be a new unstampable number replacing an old one). The list is in the thesis notes' OWED block.
- `tests/test-wiring.js` was not run on the real tree; in the worktree it needed `NODE_OPTIONS=--max-old-space-size=6144` and a restored plain store to go green (test-breaks §3.1).
- `node engine/engine_release.js verify d3d04b669e18` — whether the oldest release's manifest mismatch is real-tree or worktree-only is unverified.
- `gh api repos/willhoop/ABRA/branches/main/protection` — branch protection was not queried.
- Whether the live bot is running is not determinable from files (OPS §5).
- ROADMAP #547's on-disk receipt (the pool cache's `store_dir`) was not located (MEASURE §5).
- `engine/next_regulation.js` was not re-run on the day its own comment says a new regulation is announced (OPS §6).

The census was not pinned. The gate run's receipt reads `steering.pinned false`, `input_read_from data/mechanics-census.json`, `matches_live true` at run time — and the suite rewrote that census at 16:21:25Z, so `matches_live` would now read false. Because the census is credited-only for the differential (it does not select the sample) this does not void the run; the receipt no longer matches the live file and the re-run must carry `--census data/verification/census-pin-<id>.json` (MEASURE §2, §12).

The side-selection claim is unconfirmed. The code review reports `engine/side_selection_census.js` at **exit 1: undeclared: 88 ratchet 78 >> ROSE**. The coordinator ran it and observed it print UNCLASSIFIED rows (42751, 43091, ...) and **exit 0**. MEASURE's PENDING-WIRE entry cites `undeclared 84` measured 2026-09-04. Three numbers, two exit codes, one instrument that nothing runs (`PENDING_WIRE`, `tests/run-all.js:542`). Flagged, not resolved.

The exit-2 disagreement, reconciled as far as it can be. The code review says 44 of 144 discovered tests carry an `exit(2)` path and that `tests/run-all.js` reads exit 2 as SKIP, so a COULD-NOT-STAGE fixture launders into a green suite. MEASURE's run reports **0 skipped of 177**. Both are true: `tests/run-all.js:770-777` reads exit 2 as SKIP **by design** ("EXIT 2 MEANS 'I COULD NOT RUN', NOT 'I FAILED'"; coordinator read the lines) and `:819` exits 0 on skips; on this run no test took that path. The hazard is latent, not manifest today. It remains a defect: the exit-code vocabulary is decided in three files with three meanings (finding 15), and a fixture that stops staging tomorrow turns nothing red.

The test-wiring disagreement, resolved. The test-breaks pass says `tests/test-wiring.js` is "not referenced anywhere in `tests/run-all.js`" (`grep -n test-wiring` → nothing). MEASURE lists it red in the run. Both are literally true: `tests/run-all.js:51-52` DISCOVERS every `tests/test-*.js` by `glob ( fs.readdirSync(D('tests')).filter(f => /^test-.*\.(js|py)$/.test(f)) )`, so the file is run without being named. It ran, and it is red (10 NOT WIRED) — MEASURE attributes that to the MAG weights refusing to load ("consistent with, not proven"). Re-run the day the weights load.

Leaf coverage — three figures, one current. The ENGINE brief's premise was "compared 33 / declared 4 / neither 43 of 80" — that is the 2026-08-28 figure `engine/coverage.js` quotes in its own header (lines 10–11) as the reason the file exists. ENGINE's read-only probe today (`tests/probe_uncompared_leaves.js`): **80 leaves, COMPARED 54, DECLARED 6, NEITHER 20** — 18 with an authority duration of 1 ended in the residual, 2 self-removing, 0 with a clock of 2+. MEASURE (`node engine/coverage.js`): **54 compared — the ceiling — 6 declared uncomparable, 18 duration-1, 2 removed inside their own action, 0 widenable**. ENGINE's 20 is MEASURE's 18 + 2; the two agree, and **54 / 6 / 20 of 80 is current**. The brief's 33/4/43 is stale. The six DECLARED leaves are what a reader should know — Unburden's volatile is declared uncomparable, which is why register row #535 cannot show up on the bar.

Not measured: the share of living-document paragraphs the QUALIFIED exemption removes from clause 3b(b) — the number that decides whether finding 3 is a corner or a hole (test-breaks §4); whether WAR's gradient descent converges at low λ ; the sign and magnitude of the noisy-OR error in the role model; the full pass over docs/ENGINE.md's 167 strong-form causal lines.

PDF toolchain. `python -c "import weasyprint"` fails (OSError: cannot load library 'libgobject-2.0-0'). `build/omnibus.py` fell back to headless Chrome ("weasyprint unavailable ... trying headless Chrome; rendered with chrome.exe") and produced a 55 KB probe and the 551 KB thesis PDF. The route works; it is not the documented one.

Agents that could not write. The OPS agent has no write tool; its report was saved to disk by the coordinator.

What the fix pass itself surfaced, and left owed (each with the report that carries the command):

- The recovered store is larger than several readers assumed. Six scripts died at the default heap and now declare one (2026-09-09-fix-heap-declarations.md); any OTHER script that reads a store whole with `readFileSync(...).split('\n')` and has no declaration will die the same way the first time it runs on this store. Streaming the read is the durable fix and is owed.
- `data/live.js` is stale against the recovered store: the OPS ledger's generated block reads 81,269 ladder games where the store holds 92,379. The ingest workflow regenerates it on its next run; until then the ledger prints the pre-recovery count (2026-09-09-fix-rows.md).
- The shard-superset guard (`build/compress-stores.js --check`) exists and is red on a replay of the cutover, but nothing in `.github/workflows/ingest.yml` calls it yet. Until it is wired, the class it catches can recur unattended (2026-09-09-fix-store-recovery.md).
- `data/smogon-priors.observed.json` must be committed with this release or the cron's first `git add` exits 128 on 2026-09-11 (2026-09-09-fix-smogon-cron.md). It is staged in this commit.
- Four retracted figures (0.7048 , 52.4% , 0.687 , 1280 / 1267) could not be entered in the derived retraction registry without accusing unrelated cells in two ledgers; phrase-scoped RETRACTED entries are proposed in 2026-09-09-fix-retractions.md §D and not written.
- The record-keeping agent exited once mid-restamp with its background `status.js --write` still running, and the coordinator's concurrent suite run had to be stopped to keep a writer from sitting beside a measurement. The suite was re-run clean afterwards; the figure above is from the clean run.
- Register rows #551 (the three void games are the three board-parted games, ENGINE) and #552 (the docs gate is blind to a wrong headline, MEASURE) now exist, so `engine/open_work.js` prints both.

The rules

A system of rules across the whole project, ranked by blast radius. Each carries the ORIGINATING FAILURE (a real event with a receipt), the ENFORCING CHECK (a named test, script or hook, or "NONE — this is a preference until a check exists"), and LITERATURE (only from the list the coordinator confirmed; otherwise "no literature found"). They are written against the failure shapes this repository already names about itself: a silent default looks exactly like a working feature (LESSONS §1); every derivation over-matches on the first try (§4); a check nobody acts on is not a check; prose outlives what it described.

R1. A store's id set may not shrink across a commit

- **RULE.** For every tracked store, `ids(HEAD)  $\supseteq$  ids(HEAD~1)` . A cutover that changes the store's FORM compares against the previously TRACKED form, never against a local file.
- **ORIGINATING FAILURE.** Finding 1: 11,110 + 4,752 ids dropped at 18432bcb under a row saying "Supersedes. Nothing." The workflow's shrink guard counted lines of the shard-restored store, so a short shard set became the floor. Incident #4 (71de227c , -169 ids) is the only other shrink in the ladder history and was never noticed either.
- **ENFORCING CHECK.** At draft time NONE: `build/compress-stores.js --check` compared to the local file (forensics §1c). **Since the fix pass (§3.1)** `--check` asserts the shards $\supseteq$ the ids tracked at HEAD~1 in either form, shown red on a replay of the cutover. It is run by `tests/test-workflow-paths.js` only; the ingest workflow and the pre-commit hook do not call it yet (§8 item 2).
- **LITERATURE.** Sandve, Nekrutenko, Taylor, Hovig (2013), "Ten Simple Rules for Reproducible Computational Research", PLoS Comput Biol — version everything, record every result's provenance.

R2. A denominator may not exclude the games that failed the test

- **RULE.** Any ratio published as a correctness claim states its population as "N usable, M voided, of which K failed the test", with K derived. A void rule that correlates with the failure it is meant to count is a selection, and the artifact says so beside the number.
- **ORIGINATING FAILURE.** Finding 2: all three void games parted a board (`mid_void.void_game_tags[].board_parted_at_turn = 4, 6, 3`); "O of 958" was printed and repeated for eleven batches (session-close) without the three beside it.
- **ENFORCING CHECK.** NONE. `engine/quarantine.js` reads `state.games` less `state.games_board_never_diverged`; the `by_cause_totals.games_board_material 3` is in the same artifact and nothing compares the two. Owed: a clause in the board-material row that prints `void_games` and `by_reason_detail['low-identity'].diverged` beside the bar and refuses to print the bar alone when the second is non-zero and unattributed.
- **LITERATURE.** Gelman & Loken (2014), "The Statistical Crisis in Science", American Scientist — the garden of forking paths: an exclusion rule chosen after seeing the data is a degree of freedom.

R3. A figure gate compares the value at the claim, not membership in the artifact

- **RULE.** A living-document figure that cites an artifact is checked against the artifact FIELD the claim names, at the claim's precision. A word-list exemption applies to the sentence carrying the word, never to the paragraph.
- **ORIGINATING FAILURE.** Finding 3: 27 of 961 → 41 of 961 passed byte-identically (`engine/docs_scan.js:626 QUALIFIED` skips the paragraph; `:496-499 artifactHas` is set membership). The start skill already records "231 live of 232 probed" counted as traceable because 231 appears inside a 308 KB census" — the same class, still live.
- **ENFORCING CHECK.** `tests/test-docs-current.js` clause 3b(b) exists and was shown NOT CAUGHT on the mutation. Until it is rewritten it is a preference. Owed: the H1/H2/H4 mutations from the test-breaks report as the red arm of the rewritten clause.
- **LITERATURE.** DeMillo, Lipton, Sayward (1978), "Hints on Test Data Selection", IEEE Computer; Jia & Harman (2011), "An Analysis and Survey of the Development of Mutation Testing", IEEE TSE — a test that survives the mutation it exists to catch is not a test.

R4. A release cited by a published figure is tracked or re-cuttable

- **RULE.** The moment an artifact that a living document cites stamps a release id, that release's directory is `git add -f 'd` in the same commit. A release that cannot be opened from a clone is not evidence.
- **ORIGINATING FAILURE.** Finding 4: `b730e44f3314` stamped by all five gate artifacts, 0 tracked files; the `.gitignore` block (`:167-173`) describes the rule and nothing does it. LESSONS §12 already recorded 168 of 200 releases stranded by a reader that kept moving.
- **ENFORCING CHECK.** NONE for tracking. `tests/test-artifact-rerunnable.js` ratchets openability for releases on disk, not presence in git. Owed: a clause in `engine/provenance.js` (or the pre-commit hook) that fails when a living document cites an artifact whose `engine_release` is not in `git ls-files data/releases`.
- **LITERATURE.** Peng (2011), "Reproducible Research in Computational Science", Science 334:1226 — code and data available at the version that produced the result; Buneman, Khanna, Tan (2001), "Why and Where: A Characterization of Data Provenance", ICDT.

R5. No scheduled job writes a frozen source

- **RULE.** A workflow may write a `.observed` table beside a frozen SOURCE; promotion into the SOURCE is a human commit with a notes row and a release re-cut.
- **ORIGINATING FAILURE.** Finding 5: `smogon-stats.yml` commits `data/smogon-priors.json` (a SOURCES entry, `engine_release.js:150`) on a cron with a fresh `generated` date every run; three bot commits already (`441196b8`, `89d86b93`, `c1e2ec2f`). `ingest.yml:236-252` fixed the identical failure for `move-priors.json` — "every write here minted a NEW ENGINE and withheld every artifact measured before it" — and the lesson did not reach the second workflow.

- **ENFORCING CHECK.** NONE as a rule; the one offending workflow was fixed by hand (§3.2) and `tests/test-workflow-paths.js` now asserts it stages the `.observed` twin. Owed: the general clause — no `git add` line in any workflow names a path in `engine_release.js` SOURCES.
- **LITERATURE.** Sculley et al. (2015), "Hidden Technical Debt in Machine Learning Systems", NeurIPS — undeclared consumers and unstable data dependencies.

R6. A red test is fixed in the session that saw it, or waived by Will by name

- **RULE.** Unchanged from CLAUDE.md. There is no third state. A release is cut over zero unwaived reds.
- **ORIGINATING FAILURE.** Finding 6: 27 reds, none waived, the session-close reported three; the docs-currency gate "red on every run and reported as one of the two known failures" for two days (CLAUDE.md, 2026-07-30); `engine/provenance.js` "red for weeks and read as expected" (MEASURE §1.B).
- **ENFORCING CHECK.** `tests/run-all.js` exits 1 on any fail; `tests.yml` runs it on every push. What is missing is the WAIVER as data: a named-waiver file the runner reads, so that "waived by Will" is a row and not a memory. Owed.
- **LITERATURE.** Fowler, "Continuous Integration" (martinfowler.com) — fix broken builds immediately; a broken build is the top priority.

R7. A causal claim in a living document names its control or is marked hypothesis

- **RULE.** "X caused Y" in a living document carries the knob, probe or reproduction that showed it, or the word HYPOTHESIS. The mechanism is a claim like a number is.
- **ORIGINATING FAILURE.** Finding 7: CLAUDE.md:357 "confirmed cause" against CHANGELOG 3.23.0's counter-example and the forensics table's three no-driver duplications; 9 hypotheses stated as fact. The retraction registry cannot see a mechanism because there is no number in it to strike.
- **ENFORCING CHECK.** NONE — this is a preference until a check exists. The cheapest instrument: a `docs_scan` clause that flags a strong-form causal phrase (`confirmed cause|root cause|the cause was|which is why`) in a living document with no `knob|probe|reproduced|HYPOTHESIS` token in the same paragraph, ratcheted from today's count.
- **LITERATURE.** no literature found.

R8. A living document may not state the opposite of its artifact's verdict field

- **RULE.** Where an artifact carries a `verdict`, `supported`, or a named CI, the sentence that cites it is rewritten from that field; a sentence that contradicts it is a retraction owed NOW, not at the next major.
- **ORIGINATING FAILURE.** Finding 8: `war.json:verdict` "WORSE THAN A COIN" vs "barely clears it"; `slowing-eval.json:top_nontransitive_cycle.supported = false` vs a cycle paragraph naming a different cycle; `chomp-ev.json:selection_audit.excluded_mean_turns null` vs four numbers in the paper.
- **ENFORCING CHECK.** NONE for verdict strings; R3's rewritten clause covers the numbers. Owed: a `docs_scan` clause that, for every cited artifact carrying a `verdict` or `supported` field, checks the citing sentence for its negation words — small, bounded, and it would have caught all three.
- **LITERATURE.** Eurostat, ESS Guidelines on Revision Policy (2013), KS-RA-13-016 — a revision is documented with its reason and its impact; it is not silently overwritten.

R9. Only one publisher, and every publisher is in the table

- **RULE.** The WHO MAY WRITE table lists every agent that can land a commit, with its cadence and what it touches. A second Claude Code session, a worktree, and a bot are publishers.
- **ORIGINATING FAILURE.** Finding 12: `abra-bot` at 50 of the last 300 commits, absent from the table; a peer session running at review start; another agent's worktree walked as repo source by `test-fixture-legality`; the merge-driver binding per-clone and untested.
- **ENFORCING CHECK.** Partial: git's fast-forward rejection and `concurrency: group: ingest` serialise the bot; `/start` asks the session to run `ListAgents` (convention). Owed: `engine/orient.js` fails when `git config merge.jsonl-store.driver` is

empty; a lock file written by `/start` and read by the pre-commit hook; every repo-wide sweep excludes `.claude/worktrees`.

- **LITERATURE.** no literature found.

R10. Measure before asserting — a gate existing is not evidence it fires; mutation-test it

- **RULE.** A check is trusted only after it has been shown red on a deliberate break of the thing it protects, and the break is recorded in the check's own header. A check whose red arm has never been shown is a preference wearing a test's name.
- **ORIGINATING FAILURE.** The test-breaks pass: 12 of 13 CAUGHT, 1 NOT CAUGHT (`test-docs-current`, finding 3), two blind spots inside CAUGHT tests (`test-mc-key` skips any line beginning with `/*`; `test-sprt-arm-sign`'s spelling clause stayed green on an inverted sign). 171 of 366 test files have no documented red arm; 45 of 237 engine knobs are named in no test. The memory file records five CLAUDE.md violations in one session, all this shape.
- **ENFORCING CHECK.** `tests/test-mutation-coverage.js` (red today: "the planted-stub gate catches both stubs (0/2 caught)", artifact on release `6fb9ebd3b704`); `tests/mutation_harness.js` and `data/mutation-coverage.json` (last generated 2026-08-22). The instrument exists and is stale.
- **LITERATURE.** DeMillo, Lipton, Sayward (1978); Jia & Harman (2011); Luo, Hariri, Eloussi, Marinov (2014), "An Empirical Analysis of Flaky Tests", FSE — for the priority arm that raced and the worktree reds that were not findings.

R11. A measurement pins the engine, the pool, the census, and the flags — and the receipt names the path it opened

- **RULE.** Every differential run records `--release`, `--team-store`, `--census`, `--games`, `--turns`, the arm and the steering policy, and every digest in the receipt is taken from the file the run actually opened.
- **ORIGINATING FAILURE.** CLAUDE.md's own record: 995 vs 982 games an hour apart; 777 vs 961 from an unwritten `--games`; ROADMAP #547's receipt stamped from the live paths. This session: the gate run carried no census pin and a test rewrote the census under it (finding 15); a frozen release still opens the live bo3 store through `set_priors.js` (finding 15).
- **ENFORCING CHECK.** `engine/provenance.js` (content digests; `void: true` honoured); `engine/arms_comparable.js` for a pair. NONE for "the census was pinned" — the artifact records `steering.pinned false` and nothing refuses it. Owed: the differential refuses `--write` without a census pin when the census steers the run.
- **LITERATURE.** Sandve et al. (2013); Sculley et al. (2015).

R12. State is printed, never typed; a count in prose is stale the day it is written

- **RULE.** Unchanged from CLAUDE.md, restated because it was broken again this session: "twenty-five" SOURCES (27), "hourly" ingest (six-hourly), "23 tags with no consumer" (11), "43 uncomparing leaves" (20), "157 of 500 moves part" (0).
- **ORIGINATING FAILURE.** The fourteen handoffs; the ban list of four; and today, eleven premise corrections in the ENGINE brief alone (ENGINE §0).
- **ENFORCING CHECK.** `engine/status.js`, `engine/open_work.js`, `engine/where.js`, `engine/orient.js` — for STATE. NONE for counts typed into CLAUDE.md prose; the file says to read them from `SOURCES`, and readers do not.
- **LITERATURE.** no literature found.

R13. A published figure states what it does NOT cover, in the same sentence

- **RULE.** The one sentence 6.0.0 must print (MEASURE §3) is the template: population, pins, what was compared, and then the named exclusions — voided games, uncomparing leaves, fields not compared, mechanics nobody brought, real EV spreads (0 of 17,536 bodies carry one).
- **ORIGINATING FAILURE.** "BOARD-MATERIAL 0 of 958" quoted for eleven batches as if it were "no board difference occurred"; the session-close's own last section says it means "none survived to a compared boundary", measured at 2 of 19 narration windows.
- **ENFORCING CHECK.** NONE — the sentence exists in no living document yet (`docs/RUNNING-NOTES.md`:91 owes it to the next major).
- **LITERATURE.** Eurostat ESS (2013) Item 3.4 — documentation of the reasons, impact, and comparison for a revision.

Pre-6.0.0 list

Every item that should be addressed before 6.0.0, ranked, with owner, severity, and the one command that proves it done.

Decisions Will has already made (do not re-ask): the MAG refit is sequenced AFTER 6.0.0 (never restamp `data/policy-weights.json`; the FEATURE SEMANTICS print is correct withholding); WEB is paused (the four web tests and the two quarantine DRIFTS are the owner's, and get a named waiver); the narration clause is its own gate, not on the board-material critical path (2026-08-22); rank by the pinned pool, the roster and census carry the obscure tail (2026-08-23); documents fold in at the major, a notes row every change (2026-09-06); ledgers get no PDF.

#	item	owner	severity	proves it done
1	~~Restore the 11,110 ladder + 4,752 bo3 games to the tracked shards; supersede the "Supersedes. Nothing." row; CHANGELOG entry for the sharding (none exists)~~ DONE 2026-09-09 (§3.1): <code>comm -23 → o</code> ladder, o bo3, coordinator re-ran; the row and the CHANGELOG entry land with the record-keeping pass (§3.5)	OPS	~~BLOCKS~~ closed	<code>comm -23 <d2a418a5 ids> <shard ids></code> <code>\ wc -l → o</code> , for ladder and bo3 — read o / o
2	~~Add <code>ids(HEAD shards) ⊇ ids(HEAD~1 shards)</code> to <code>compress-stores.js --check</code> ~~ DONE 2026-09-09 (§3.1: red on a one-row break, red on a replay of 18432bcb, green on the tree in 8.7 s). STILL OWED: run it in <code>ingest.yml</code> after the shard step and in the hook on any commit touching <code>data/parsed/</code> ; the runner's shrink guard still measures against itself	MEASURE / OPS	SHOULD FIX (the wiring)	<code>grep -n 'compress-stores.js --check' .github/workflows/ingest.yml → a hit</code> after the shard step
3	Attribute the three void games (charge-move release target; Future Sight vs side-end order; post-switch secondary) — one probe each; print "o of 958 usable, 3 voided, N engine"	ENGINE	BLOCKS	the three probes exit o in both engines, and <code>by_reason_detail['low-identity'].diverged</code> is attributed in the artifact
4	~~Stop the smogon cron writing a frozen source before 2026-09-11 07:00 UTC (<code>.observed</code> table; drop the SOURCE from <code>git add</code>)~~ DONE 2026-09-09 (§3.2): workflow-only, drift <code>b730e44f3314 → NO-DRIFT</code> ; <code>data/smogon-priors.observed.json</code> must be added with the commit	OPS	~~BLOCKS~~ closed	<code>node engine/smogon_priors.js && git diff --stat -- data/smogon-priors.json → empty</code> — the same-month run moved 2 lines before the fix; the workflow now restores the file in the same step
5	<code>git add -f data/releases/b730e44f3314</code> ; a provenance clause failing on a cited untracked release	MEASURE	BLOCKS	<code>git ls-files data/releases/b730e44f3314 \ wc -l > o</code>
6	Rewrite <code>docs_scan</code> clause 3b(b): value at the named field; qualifier exempts the sentence	MEASURE	BLOCKS	the H1/H2/H4 mutations from the test-breaks report turn <code>tests/test-docs-current.js</code> red
7	Fix or waive-by-name every one of the 27 reds; a named-waiver file the runner reads	all; Will for group A	BLOCKS	<code>cmd /c tools\lownode.cmd tests\run-all.js → o</code> failed, waivers printed by name
8	~~Strike the five refuted causal claims (<code>CLAUDE.md:357</code> , <code>ARCHITECTURE.md:42-43</code> , <code>SEARCH.md:2506</code> , <code>ENGINE.md:404/410-411/419-420</code> , <code>CLAUDE.md:577-579</code> marker)~~ DONE 2026-09-09 (§3.3, 8 locations; <code>test-docs-current 33/o</code>). STILL OWED: <code>push-all.bat:9-10</code> , <code>.gitattributes:9-11</code> , <code>ingest.yml:355-357</code> — live files, not living documents, still naming the driver as THE mechanism	MEASURE / ENGINE / OPS	~~BLOCKS (R4)~~ closed; SHOULD FIX (the live files)	<code>grep -n "confirmed cause\ ONE IN SIX\ publishes on a timer" CLAUDE.md docs/*.md → nothing</code>
9	~~Rewrite the three white-paper sections from their artifacts' verdict fields (WAR :1968, :1986; NMF :1978-1983; CHOMP-EV :1691-1699) — retractions do not wait~~ DONE as a retraction 2026-09-09 (§3.3, §4: 22 figures across 21 locations, every replacement	MEASURE	~~BLOCKS~~ closed for the sentences; the analyses stay owed	<code>node tests/test-docs-current.js 33/o</code> and <code>node engine/docs_scan.js -- quarantine</code> unchanged at 23 — both read ; the sentences now quote <code>war.json:verdict</code>

#	item	owner	severity	proves it done
	read from the artifact). The re-analyses are not done — T1–T5 below			
10	Write the 6.0.0 notes row with **Basis.** CHANGED — ... ; version the two :366 / :444 rows under it; say PARTIAL LIFT (24 stay) in words	MEASURE	BLOCKS	node tests/test-docs-current.js clause 5d passes on a 6.0.0 entry
11	State the roster's compared set wherever it is cited: 139 of 202 abilities, 142 of 148 items, 487 of 500 moves	ENGINE	BLOCKS (wording)	node -e over data/roster.*.json verdict counts matches the sentence
12	Infiltrator through the four screens — four census probes	ENGINE	BLOCKS	node tests/test-mechanics.js and all_mechanics_fire.js --kind all --write --release <id> show four new rows FIRED-AND-BOARDS-MATCH
13	The one-sentence bound on "o of 958" (MEASURE §3) into every living document	MEASURE	SHOULD FIX	grep -c "3 were voided" docs/ABRA-whitepaper.md docs/SUMMARY.md docs/MODELS.md ≥ 1 each
14	Substitute family: step-o doll slot, then the outer arrival loop with #500/#511/#361, bar checked between stages	ENGINE	SHOULD FIX	tests/test-resolution-order.js 's KNOWN-OPEN arm closes; differential re-run board-material unchanged
15	Spicy Spray order + correct the two false comments at :40250 , :40323 ; Trick's guard → itemRefusesTake ; perish \ upkeep \ drain; post-KO switch order; Unburden as a volatile	ENGINE	SHOULD FIX	narration causes fall by five on the re-run at the same pins
16	Compound Eyes probe; fix the Imprison/Memento pass, move 1 throw	ENGINE	SHOULD FIX	tests/roster.js --stage abilities / - -stage moves --write → the rows leave COULD-NOT-STAGE
17	engine/mew.js to STAY; major_readiness predicate follows one require hop	MEASURE	SHOULD FIX	node engine/major_readiness.js lists mew.js under STAY
18	Pin the census on the re-run; stop tests/test-arm-steering.js writing data/mechanics-census.json	MEASURE	SHOULD FIX	artifact steering.pinned true ; git status clean after the suite
19	Consume commit_matches : the differential's preflight fails on drift; engine_release.js cut --fail-on-drift	ENGINE	SHOULD FIX	move the checkout one commit, the differential refuses
20	ABRA-HEAP headers on engine/engine_release.js , engine/mew.js , tests/test-wiring.js ; run the release census at 6144	MEASURE	SHOULD FIX	node --max-old-space-size=6144 engine/engine_release.js census finishes
21	One exit-code classifier imported by run-all , register_reality , wire_ladder ; a COULD-NOT-STAGE exit counts as red unless ABRA-EXIT 2 CANNOT-ANSWER is printed for an environment reason	MEASURE	SHOULD FIX	a test made to exit 2 on an illegal fixture turns the suite red
22	WHO MAY WRITE: add abra-bot (six-hourly, data/only); orient.js fails on an unbound merge driver; exclude .claude/worktrees from every sweep	OPS / MEASURE	SHOULD FIX	node engine/orient.js prints the driver clause; tests/test-fixture-legality.js green with a worktree present
23	Back up the pinned pool outside the tree; --restore-parsed the local stores; refresh the legality verdict	OPS	SHOULD FIX	sha256sum data/team-pool-frozen/*.jsonl matches FROZEN.md; node tests/test-quality.js green
24	Derive MEGA_ABIL and validate_damage.js 's type chart; count the two empty catches at :5500 , :6275	ENGINE	SHOULD FIX	tests/test-effective-identity.js and engine/validate_damage.js green with the literals gone

#	item	owner	severity	proves it done
25	<code>register_reality.js --list</code> (verdicts a day old); flip the seven closed-on-tree register cells	ENGINE	SHOULD FIX	<code>node engine/open_work.js</code> shows the rows closed
26	~~Version header or baseline exemption for docs/THESIS-DEFENCE-REVIEW-2026-09-09.md and this file~~ DONE 2026-09-09 (§3.3: both added to <code>unversioned_exempt</code> in <code>data/docs-currency-baseline.json</code> ; clause 2b green, baseline 59 / now 59)	MEASURE	~~SHOULD FIX~~ closed	<code>node tests/test-docs-current.js</code> clause 2b green — read green, 33/o
27	Residual trio: say the mutual-residual-KO corner is unmeasured; 16 zero-census tags; damage-differential multi-hit band; self-play 89 dups; the nine CRLF sources; <code>where.js --artifacts</code> heuristic	ENGINE / MEASURE / OPS	DEFER WITH REASON	stated in the 6.0.0 release note

The thesis's twelve (sub-block; owner MEASURE unless stated; from `docs/THESIS-DEFENCE-REVIEW-2026-09-09.md` "Required to reach a pass")

#	analysis	size	proves it done
T1	NMF: read the rank from <code>nmf-rank-selection.json:most_reproducible.rank</code> or state rank 6 ships below null; delete "error 0.53"; replace the archetype table; re-run <code>nmf_rank.py</code> at ≥50 pairs with cophenetic correlation	hours + a run	<code>data/nmf-roles.json:archetype_rank</code> equals the criterion's, or the three documents say why not
T2	WAR: three-way id-hash split; select λ on validation; report test once; record split, seed, grid, store digest	a day	<code>data/war.json</code> carries <code>split</code> , <code>seed</code> , <code>store_digest</code>
T3	CHOMP-EV: populate <code>audExcl</code> below the qualify test; replace 1,205 / 6.5 / 6.08 / 1280 / 1267 with fields	a day	<code>chomp-ev.json:selection_audit.n_excluded_human > 0</code>
T4	SLOWKING: replicates at the point estimate's iteration count; remove "substantially less exploitable"; rewrite §4.5 from <code>supported</code> and the CI	a day	<code>slowing-eval.json</code> intervals contain their point estimates
T5	§1 ceiling: build <code>engine/skill_variance.py</code> → <code>data/skill-variance.json</code> or withdraw 0.687 / 0.693 / 52.4%	real work	the artifact exists, or the figures are gone from §1 and <code>PUBLICATION.md</code>
T6	XATU: reconcile :1676-1678 and <code>MODELS.md:1405</code> with <code>data/policy-eval.json</code>	hours	the sentence quotes 0.2979 / 0.6564 / 2.6353 or names the artifact carrying 35.9 / 71.6
T7	Noisy-OR: state the assumption; measure direct vs noisy-OR P(team has role) per role; publish the signed gap	real work	an artifact field per role
T8	GURU in the white paper: the family (53 cells, 3 decisive, 2.65 expected); 0.7122 → 0.7124; in-sample stated; silhouette 0.126 beside K	hours	§4.1 quotes <code>predictive_test</code> and the family
T9	Roles count: 26 → derived (52 signals / 46 columns)	hours	the number is generated in the doc build
T10	Bring-rate bias magnitude from one pinned pass into <code>bring-bias.json</code> or <code>quality-filter.json</code> ; cite at :1707	a day	both operands (kept/dropped, mean turns, per-species shift) in the artifact
T11	Reproducibility (#288): store digest + game-id-set digest + split + seed in every store-derived artifact; then re-run the store layer once on filter 1.3.0	real work; makes the rest checkable	<code>data/provenance-stamp.json:mtime_only</code> falls by the twelve artifacts named
T12	Citations: RAPM (Rosenbaum 2004 / Sill 2010), Pearl 1988, Hart & Mas-Colell 2000; resolve two VGC-Bench titles; fix or remove the 2603 arXiv id; mark Geng 2016 and arXiv 2304.08272 as motivation	hours	the reference list resolves

The thesis defence

The third defence returned **MAJOR REVISIONS**. The blocking item from July — the NMF rank — was not merely left open; the artifact was regenerated at the undefended rank four days after that committee sat (`data/nmf-roles.json:generated` 2026-08-04, rank 6, excess over null -0.107), and the white paper still justifies it with a number its own script cannot produce. Beyond that, at least twelve figures in the white paper disagree with the artifact the sentence cites, one "measured" selection audit structurally measured zero games, a regularisation parameter was selected on the partition its score is reported on, and confidence intervals the generator annotates as "unusable" are quoted as evidence. Not one store-derived artifact records which games it read. The doctoral claim itself — exploitability — is unmeasured and the paper says so. The measurement culture the July committees praised is real and has improved; the defects are in exactly that layer, which is why the grade is what it is. Twelve required analyses, five of them hours, are listed above and in full at [docs/THESIS-DEFENCE-REVIEW-2026-09-09.md](#); every command and raw output is in `docs/_reports/2026-09-09-thesis-defence-notes.md`.

Reviewed against commit `d6789951`. Findings 1, 2, 4, 5 and the union-driver claim were confirmed by the coordinator's own commands; everything else is cited to the division report that carries it. Nothing under `engine/`, `tests/` or `data/` was edited by the review draft; the three fix passes it records (§3) edited `build/compress-stores.js`, `engine/dedupe_store.py`, `data/parsed/` (additions only) and `data/docs-currency-baseline.json`, and nothing under `tests/`. This file is a dated record.